



## OPEN A hybrid deep learning algorithm for ECG-based heart disease classification

Sa. I. Ibrahim

Automated electrocardiogram (ECG) classification for heart disease detection encounters significant challenges, such as patient variability, inter-patient differences in heart signals, and data leakage when information from the same patient is present in both training and test sets. To mitigate these challenges, a Hybrid 1D CNN-BiLSTM (one-dimensional convolutional and bidirectional long short-term memory) deep learning framework is introduced, employing rigorous patient-wise validation on the MIT-BIH Arrhythmia Database. Patient-wise data splitting (34 records for training, 14 records for testing) ensures that each patient's data is confined to a single set, effectively preventing data leakage. Under these conditions, the proposed Hybrid with GAN model achieves 91.69% accuracy (95% confidence interval: 91.41–91.99%), with a macro F1-score of  $0.547 \pm 0.374$ . Conditional GAN augmentation significantly improves minority class performance: Supraventricular recall increases from 9.2% to 20.9% (+127.2%), and Ventricular recall improves from 77.7% to 80.9% (+4.1%). Statistical significance testing using McNemar's test demonstrates the superiority of Hybrid with GAN over the non-augmented hybrid model ( $\chi^2 = 12.41$ ,  $p = 0.0004$ ). The observed 7.31% performance gap between beat-wise (99.0%) and patient-wise (91.69%) evaluation highlights the critical importance of patient-wise validation for clinically realistic performance estimates. Overall, this framework provides a rigorous patient-wise evaluation protocol for automated ECG classification, with patient-wise results for a GAN-augmented hybrid model on the full 5-class MIT-BIH dataset.

**Keywords** Deep learning, Heart disease, 1D-CNN, BiLSTM, ECG, Patient-wise validation, GAN augmentation

### Abbreviations

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| AG          | Average gradient                                                              |
| AAMI        | Association for the advancement of medical instrumentation                    |
| AI          | Artificial intelligence                                                       |
| CI          | Confidence interval                                                           |
| CNN         | Convolutional neural networks                                                 |
| CVD         | Cardiovascular disease                                                        |
| Conv-LSTM   | Convolutional LSTM                                                            |
| DL          | Deep learning                                                                 |
| DT          | Decision tree                                                                 |
| DTRF        | Decision tree-random forest                                                   |
| ECG         | Electrocardiogram                                                             |
| ECG-LM      | ECG-language model                                                            |
| FN          | False negative                                                                |
| FP          | False positive                                                                |
| GAN-1D-CNN  | Generative adversarial network—a one-dimensional convolutional neural network |
| GAN-Bi-LSTM | Generative adversarial network—bi-directional long short-term memory          |
| HD          | Heart disease                                                                 |
| IQR         | Inter quartile range                                                          |
| K-NN        | K-nearest neighbors                                                           |
| LR          | Logistic regression                                                           |
| LSTM        | Long short-term memory                                                        |

Information Systems Department, Faculty Of Computers and Informatics, Suez Canal University, Ismailia, Egypt.  
email: sara\_hassan@ci.suez.edu.eg

|         |                                                              |
|---------|--------------------------------------------------------------|
| MAE     | Mean absolute error                                          |
| MIT-BIH | Massachusetts institute of technology - beth israel hospital |
| ML      | Machine learning                                             |
| 1D-CNN  | One-dimensional CNN                                          |
| PCA     | Principal component analysis                                 |
| ResNet  | Residual network                                             |
| RF      | Random forest                                                |
| RMSE    | Root mean square error                                       |
| RNN     | Recurrent neural networks                                    |
| SGB     | Stochastic gradient boosting                                 |
| SVM     | Support vector machines                                      |
| TN      | True negative                                                |
| TP      | True positive                                                |
| WHO     | World Health Organization                                    |

Heart disease (HD) is considered one of the main reasons for deaths around the world. HD can be any kind of disorder that the heart affected with. These disorders can be blood vessel disease, infection, and, etc. It is also a key problem in global health worldwide. Early and accurate detection of HD is critical for efficient treatment of patients and preventing the danger of improving this disease. Traditional diagnosis methods consume more time, and very expensive. HD mainly dies around 17.9 million people die from heart disease each year based on the World Health Organization (WHO)<sup>1</sup>.

Artificial Intelligence (AI) offers computational model that may execute tasks which are typical for human intelligence and decision-making. Machine learning (ML) is a subdomain of AI. ML mainly involves creating computer models to learn from data and predict or make optimal decisions. It aims to allow systems to perform a specific task and enhance performance. ML is a very useful technology in the healthcare sector, which produces timely computation models when dealing with big medical datasets<sup>2</sup>. Deep learning (DL) is a subset of ML that can do complex computational models simple. DL has led to substantial breakthroughs in the healthcare sector. Technological advancements have solved complex problems, decreased expenses, and increased efficiency in healthcare.

There is several research on diagnosing and detecting HD. ML methods help diagnose and detect HD. ML methods aim to extract valuable knowledge from enormous amounts of data. Several ML methods, including decision trees and k-nearest neighbors (K-NN), have been proposed in the past and achieved middle accuracy. For early diagnosis, an automated method detects the disease at the time<sup>1</sup>. The ML methods are vital for identifying and distinguishing distinct characteristics in electrocardiogram (ECG) signals. The objective is to improve the accuracy and effectiveness of the HD diagnosis by implementing current ML methods.

An ECG is a medical test that diagnoses cardiac abnormalities by measuring the electrical activity generated by the heart. It is crucial for monitoring and diagnosing heart disease. Electrodes are placed on the chest and legs to monitor and analyze the heart's electrical activity<sup>3</sup>. Manual diagnosis presents technical difficulties due to significant variances in ECG. Recent research emphasizes the use of ML and DL methods to enhance the automatic extraction of key features for HD diagnosis. Extracting information from ECG signals is vital for accurately diagnosing HD<sup>4</sup>. Diagnosis efficiency is based on the quality of the feature extraction process.

This study investigates the use of convolutional neural networks (CNNs) and the Bidirectional Long Short Term Memory (Bi-LSTM) to study ECG signals and classify these. The main aim is to do a DL system that can aid doctors in timely and accurate diagnose to improve patient outcomes.

HD causes over 17.9 million deaths worldwide each year, and delayed detection highly affects patient outcomes. Traditional ECG focuses on clinicians' manual determination, and it's time-consuming and open to human error. While automated ECG classification methods are available, many are not robust across many patients. Traditional ML algorithms trouble noise and class imbalance in ECG signals, whereas successful ECG classification requires determining the spatial and temporal features. The latest developments in ML and DL such as 1D-CNN and Bi-LSTM present new prospects to overcome these barriers by facilitating high accuracy of complex ECG signals. 1D-CNN seeks to extract local information but frequently lacks vital patterns, whereas Bi-LSTM deals with temporal features. Therefore, the hybrid method determines all spatial and temporal features.

Recent deep learning methods have demonstrated great accuracy in ECG classification. Most studies use beat-wise splitting, which means that individual beats from the same patient randomly appeared in both the training and test sets, resulting in data leakage and inaccurate performance. No previous research has systematically examined beat-wise (optimistic) versus patient-wise (realistic) evaluation of GAN-augmented hybrid methods. This paper attempts to fill that gap, inspired by the urgent need for trustworthy, accessible methods to improve early, accurate HD detection and clinical decision-making.

The CNN, BiLSTM, and GAN were addressed separately in the prior works. The use of CNN, BiLSTM, and GAN augmentation for beat-wise and patient-wise splitting on the MIT-BIH 5-class is proposed. The proposed hybrid method with GAN augmentation outperforms other evaluated methods. The primary novelty and contributions of our work are:

- We conducted two experiments using an identical hybrid architecture: beat-wise splitting (traditional, 99.00% accuracy) and patient-wise splitting (strict, no overlap, no data leakage, 91.69% accuracy). The proposed 1D CNN-BiLSTM with GAN augmentation achieves the highest accuracy for a 5-class patient-wise evaluation on MIT-BIH. The observed 7.31% generalization gap indicates that beat-wise assessment substantially overestimates clinical performance. This study provides a systematic patient-wise evaluation of a GAN-augmented hybrid 1D CNN-BiLSTM model on the 5-class MIT-BIH dataset.

- In beat-wise splitting, the conditional GAN increased Fusion recall from 48% (1D CNN) to 77% (a 29% absolute improvement) and Supraventricular recall from 62% to 74% (a 12% absolute improvement). In patient-wise splitting, the GAN enhanced overall accuracy from 90.68% to 91.69% (a 1.01% absolute increase, McNemar  $p = 0.0004$ ), and improved ventricular recall from 77.7% to 80.9% (a 4.1% relative increase), supraventricular recall from 9.2% to 20.9% (a 127% relative increase), and fusion recall from 1.1% to 2.1% (a 90.9% relative increase).
- The proposed method applies focal loss with  $\gamma=2.0$  and  $\alpha=0.25$  (beat-wise)/ $\alpha=0.5$  (patient-wise) to address the arrhythmia class imbalance problem and enhance minority class identification.
- Our proposed GAN-augmented method produced statistically significant improvements over the non-augmented hybrid method in both assessment procedures (beat-wise:  $p < 0.05$ ; patient-wise:  $p = 0.0004$ ).
- For statistical validation, we offer 95% bootstrap confidence intervals for accuracy (for patient-wise Hybrid-GAN: [91.41%, 91.99%]), macro metrics with mean  $\pm$  standard deviation (patient-wise: macro F1 =  $0.547 \pm 0.374$ ), and McNemar's test for statistical significance.

In summary, we propose a hybrid algorithm based on the 1D-CNN method for spatial feature extractions and Bi-LSTM for temporal patterns. The proposed algorithm improves the result quality to aid doctors in disease diagnosis. The rest of the paper is organized as follows. Section 2 focuses on some previous works. Section 3 presents the proposed algorithm used in this paper. The experimental results and performance evaluation are discussed in section 4. Finally, We conclude and summarize this paper in Section 6.

## Literature survey

Researchers presented multiple ML methods for detecting, classifying, and predicating HD from ECG signals. All of these methods are tested and achieved good results. In this section, we preview and analyze some of these researches.

Romdhane et al.<sup>5</sup> suggested a CNN method with focal loss for ECG heartbeat classification, primarily to address the difficulty of class imbalance in arrhythmia detection. The focal loss function for object detection trains on challenging, misclassified samples, making it ideal for unbalanced medical datasets. This paper uses the MIT-BIH Arrhythmia Database to validate this method using the AAMI-recommended five-class (Normal, Supraventricular, Ventricular, Fusion, and Unknown). The suggested method has an overall accuracy of 98.41%.

Mehmood et al.<sup>6</sup> proposed a CNN method to predict cardiovascular disease. This method uses the UCI dataset. The results declared that the proposed method outperforms traditional methods, achieving a 97% accuracy rate.

Pandey et al.<sup>7</sup> presented the convolutional encoded feature (CEF) and Bidirectional long short-term memory (BiLSTM) network method for detecting arrhythmias from ECG data. The experiment classifies and detects arrhythmias in the MIT-BIH arrhythmia dataset. Performance was evaluated using recall, accuracy, F-scores, and precision. Results show that the BiLSTM method obtained an accuracy rate of 99.52%.

Sarra et al.<sup>8</sup> presented two DL methods for predicting HD, including a generative adversarial network-a one-dimensional convolutional neural network (GAN-1D-CNN) and a generative adversarial network-BiLSTM (GAN-BiLSTM). These two methods are tested using three distinguishable datasets: Cleveland, Comprehensive, and Statlog. The accuracy achieved for these two presented methods is 99.1% and 99.3%, respectively. The results declared that this method is very good at predicting heart disease.

Rezaee et al.<sup>9</sup> introduced a hybrid CNN-GCN framework to detect multiple types of cardiovascular disease using heart sound signals. We convert PCG signals into log-spectrograms to extract frequency-domain features. A deep GCN with sampling and prior layers then captures relationships between spectrogram clusters and disease patterns. We combine these GCN features with those from the CNN to improve the model's ability to distinguish between diseases. The model reached 99.44% accuracy on the Michigan database and 96.16% on PhysioNet/CinC 2016, showing strong performance.

Ahmed et al.<sup>10</sup> presented a one-dimensional CNN (1D-CNN) to classify cardiac arrhythmias based on ECG signals. This study uses the MIT-BIH dataset, which contains ECG signals. Precision, F1-score, accuracy, average, recall, and AUC-ROC were calculated to assess the performance of the presented method. Results denoted that the presented method performed well for fast diagnosis of cardiac arrhythmias with accuracy rates of 1.00 and 0.99 in training and testing this dataset.

Golande et al.<sup>11</sup> presented ECG-based HD classification using CNN and LSTM methods. The presented method comprises four stages: preprocessing, segmentation, feature selection, and classification. The results confirm that the presented method is less consuming when compared to other methods.

Omarov et al.<sup>12</sup> applied a new method that used ECG data to detect HD using the 3D Convolutional LSTM (Conv-LSTM) method. This method aims to mix the features from CNN and LSTM methods. The applied method aims to detect perceptive changes in the ECG data over time. This method was trained on ECG data from several patients in varied heart conditions like heart failure and arrhythmia. Results confirm that the 3D Conv-LSTM method is more effective than other traditional methods in detecting HD areas, while gaining an 88% accuracy rate.

Sudha et al.<sup>1</sup> presented a mix of CNN and LSTM to predict HD. It also uses the Support Vector Machine (SVM) to select features. Results gained 89% accuracy compared to Naive Bayes, SVM, and Decision Tree. The presented methods perform better than other compared methods.

Jawalkar et al.<sup>13</sup> applied the decision tree-random forest (DTRF) method to predict HD from ECG signals. Using stochastic gradient boosting (SGB) to train the applied method. The results declared that the DTRF method achieved higher performance than other compared methods, with an accuracy rate of 96%.

Zacarias et al.<sup>14</sup> proposed the LSTM method to forecast ECG signals. This study uses the MIT-BIH dataset. The mean absolute error (MAE) and root mean square error (RMSE) are used to evaluate the proposed method's performance. The results present that the proposed method is highly efficient for the healthcare area.

Saikumar et al.<sup>15</sup> proposed a Bi-LSTM method for classifying ECG signals into normal and abnormal patients. Evaluation demonstrates that the proposed method gains a 90% accuracy rate. The presented method improved ECG classification and was very vital in clinical diagnosis.

Singh et al.<sup>16</sup> applied supervised ML methods like decision tree (DT), SVM, Random Forest (RF), and Principal Component Analysis (PCA) to detect HD. The performance evaluation measures are used to evaluate the performance of these methods, like precision, specificity, F1-score, accuracy, recall, sensitivity, and AUC-ROC. The results declare that the proposed study assists doctors in the accurate detection of HD.

Nawaz et al.<sup>17</sup> presented SVM, RF, DT, and logistic regression (LR) methods to classify HD. The effectiveness of these methods is evaluated using a dataset of ECG images. The results declare that the SVM method ensures early HD detection and gains an accuracy rate of 84% as compared with other methods.

Zabihi et al.<sup>18</sup> proposed both ML and DL methods to classify ECG signals. This work applied both a residual network (ResNet) to extract ECG signal features and an LSTM to classify these features. Then uses the RF method in ECG signals classification. The evaluation of the proposed method was implemented on the MIT-BIH dataset. This method achieved an effective and accurate classification of ECG signals with a 99.26% accuracy rate.

Venkatesh et al.<sup>19</sup> proposed an ensemble deep network algorithm that combines 1D-CNN and BiLSTM methods for arrhythmia classification. This paper was tested on the MIT-BIH Arrhythmia Database, and the method obtained an accuracy of 98.5%, with good sensitivity and specificity across numerous arrhythmia types.

Ran et al.<sup>20</sup> introduced a multiple-feature, multiple-path parallel Deep Convolutional Neural Network (DCNN) with a better focal loss function for arrhythmia classification. The method's multi-path design collects characteristics at multiple temporal scales simultaneously, and the enhanced focal loss addresses class imbalance by automatically adjusting weights for difficult-to-classify samples. This work adds a perfect feature to the existing literature by explicitly comparing beat-wise and patient-wise evaluation methods, which is frequently overlooked. The method obtained 98.73% accuracy, 94.89% average sensitivity, and 97.42% specificity when evaluated beat-wise. However, under the more clinically realistic patient-wise evaluation, accuracy decreased to 91.22%.

Jaiswal and Mitra<sup>21</sup> proposed a CNN-BiLSTM architecture for ECG arrhythmia classification with data augmentation (time-warping and jittering) to solve the class imbalance challenge. The external evaluation of the PTB-XL dataset, in addition to the MIT-BIH database, is a significant strength of this work. On MIT-BIH, the hybrid model obtained 98.0% accuracy, 97.5% precision, 98.2% recall, and 97.8% F1-score, whereas on PTB-XL, it achieved 96.8% accuracy, 96.2% precision, 96.5% recall, and 96.3% F1.

Ye et al.<sup>22</sup> introduced a hybrid framework that combines CNN and BiLSTM networks to detect arrhythmias from ECG data. The authors used the Mish activation function to improve training robustness. The model was trained and tested using a mix of the MIT-BIH Arrhythmia Database and clinical ECG data. The experimental results indicated an accuracy of 99.52%, a sensitivity of 99.48%, and a specificity of 99.85%.

Wani et al.<sup>23</sup> presents the use of CNN, clinical data, and SHAP explanations to predict heart disease. The fusion model achieves 98.5% accuracy by combining ECG data and patient information. SHAP values identify which features have the most effect on each prediction, hence enhancing clinician trust. This study verifies the use of fusion modeling and XAI for ECG arrhythmia classification.

Adnan et al.<sup>24</sup> presents a digital twin architecture for creating a virtual replica of a patient's heart. The technology uses neural networks and medical knowledge guidelines to anticipate cardiac issues in real-time. When tested on 10,000 patients, it achieved 98.5% accuracy for predicting CVD events. This motivates our future work on patient-specific ECG model adaptation.

In<sup>25</sup> SFX-GAN reduces haze from photos by utilizing GANs and frequency-domain analysis. The model's decisions are explained using Grad-CAM heatmaps and a Concept Bottleneck Layer. It provides world-class performance (35.10 dB PSNR) with only 13.2 million parameters. This shows how GANs may be made explainable, which directly informs our ECG GAN technique.

Singh et al.<sup>26</sup> presents DiaXplain that detects diabetes using deep learning and gives easy text explanations for its conclusions. The model leverages attention processes to extract crucial risk indicators from patient data. It achieves 98.24% accuracy on 50,000 cases, with explanations approved by physicians. This demonstrates that AI can provide relevant textual explanations for medical diagnosis.

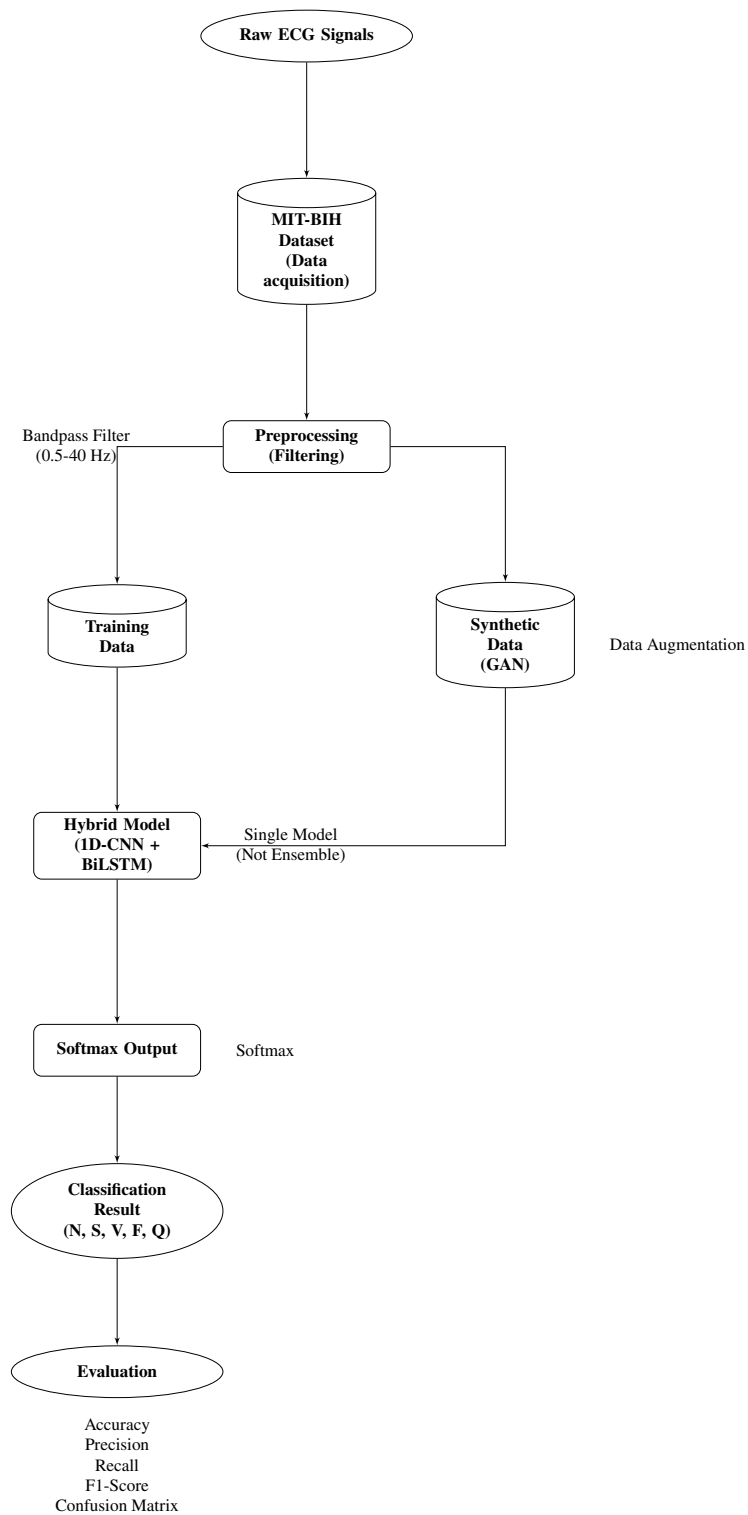
The work<sup>27</sup> combines mammograms, tissue samples, and clinical data to detect breast cancer. The fusion model achieves 98.29% accuracy and shows heatmaps of suspicious areas. Integrated gradients highlight which image regions and clinical features drive predictions. This validates that fusion models work well for medical classification tasks.

This paper looks at XAI approaches for wearable medical devices and remote patient monitoring. It discusses approaches such as SHAP, LIME, Grad-CAM, and attention processes. The study highlights problems such as balancing accuracy, interpretability, and regulatory compliance. This document provides a methodology for installing interpretable ECG models in IoMT setups<sup>28</sup>.

DeepXplainer diagnoses lung cancer in CT images using a CNN with a dual-attention approach. It offers three different explanation methods: Grad-CAM heatmaps, SHAP values, and LIME explanations. The model has a 97.43% accuracy rate and displays to radiologists exactly where worrisome nodules are located. This illustrates how several XAI approaches may be combined for medical imaging<sup>29</sup>.

## Materials and methods

This section explains the full methodology for this study. The methodology contains several steps, including data acquisition, preprocessing, Data Augmentation, model training, classification, and evaluation. Figure 1 and



**Fig. 1.** Flow diagram of the proposed Hybrid+GAN method for 5-class ECG classification.

Algorithm 1 illustrate the full methodology of the proposed study. All these steps are discussed in detail in the following sub-sections.

- 
- 1: **Input:** MIT-BIH raw ECG signals
  - 2: **Output:** Classification labels (N, S, V, F, Q)
  
  - 3: **Preprocessing:**
    - 4: a. Apply bandpass filter (0.5–40 Hz) for noise removal
    - 5: b. Apply R-peak detection
    - 6: c. Extract 187-sample window centered at R-peak
    - 7: d. Apply robust scaling using median and IQR
    - 8: e. Split data: 34 records for training, 14 for testing
  
  - 9: **Data Augmentation (Conditional GAN):**
    - 10: a. Train Conditional GAN on training data only
    - 11: b. Generate synthetic beats for minority classes (S and F)
    - 12: c. Combine synthetic data with real training data
    - 13: d. Augmented training size: 82,360 samples
  
  - 14: **Single Hybrid Model Training:**
    - 15: a. 1D-CNN for feature extraction (Conv1D + MaxPool)
    - 16: b. BiLSTM (128 units) for temporal processing
    - 17: c. Dense layer with softmax activation (5 units)
    - 18: d. Loss function: Focal loss ( $\gamma = 2.0$ ,  $\alpha = 0.5$ )
    - 19: e. Optimizer: Adam (learning rate =  $1 \times 10^{-4}$ )
  
  - 20: **Model Checkpoint and Early Stopping:**
    - 21: a. Save best weights based on validation accuracy
    - 22: b. Early stopping with patience of 10 epochs
    - 23: c. Best epoch for Hybrid+GAN: 40 (val\_acc: 98.90%)
  
  - 24: **Classification:**
    - 25: a. Generate predictions on patient-wise test set
    - 26: b. Output 5-class labels: N, S, V, F, Q
  
  - 27: **Evaluation:**
    - 28: a. Compute accuracy, precision, recall, F1-score
    - 29: b. Generate confusion matrix
    - 30: c. Calculate macro F1, MCC, and Kappa
    - 31: d. McNemar's test for statistical significance
- 

**Algorithm 1.** Proposed Hybrid+GAN algorithm.

---

### Data descriptions

The original ECG signals used in this study, data were obtained from the MIT-BIH dataset. The MIT-BIH Arrhythmia Dataset contains 48 half-hour two-lead ECG recordings sampled at 360 Hz. The recordings were taken from 47 people, including 25 males aged 32–89, and 22 women aged 23–89, at Boston's Beth Israel Hospital. Heartbeats were classified into five categories based on the Association for the Advancement of Medical Instrumentation's (AAMI) suggested standard: normal (N), supraventricular (S), ventricular (V), fusion (F), and unknown (Q). Table 1 shows the class mapping of MIT-BIH symbols to AAMI classes<sup>30</sup>.

The MIT-BIH Arrhythmia Database is a well-established benchmark for ECG-based heart disease classification, publicly available through PhysioNet at <https://physionet.org/content/mitdb/1.0.0/> (DOI: <https://doi.org/10.13026/C2F305>). The database was developed by the Massachusetts Institute of Technology and

| Class ID | AAMI classification  | MIT-BIH symbols | Number of beats | Percentage |
|----------|----------------------|-----------------|-----------------|------------|
| 0        | Normal (N)           | N, L, R, e, j   | 75,000          | 83.5%      |
| 1        | Supraventricular (S) | A, a, J, S      | 2,500           | 2.8%       |
| 2        | Ventricular (V)      | V, E            | 6,000           | 6.7%       |
| 3        | Fusion (F)           | F               | 600             | 0.7%       |
| 4        | Unknown (Q)          | /, f, Q, U      | 7,000           | 7.8%       |

**Table 1.** AAMI class mapping for MIT-BIH dataset.

---

Beth Israel Hospital in the 1970s and has been widely used in ECG classification, arrhythmia detection, signal denoising, and DL applications.

The main concern of dataset is class imbalance. The Normal class comprises 83.5% of all beats, while the Fusion class represents only 0.7% – an imbalance ratio of approximately 119:1. This extreme imbalance poses significant challenges for deep learning models, which tend to bias predictions toward majority classes. To solve this issue, we obtain beat-wise, and patient-wise splitting with and without GAN augmentation.

The two ECG leads found in each record are named MLII (modified limb lead II) and V1. MLII is a standard bipolar limb lead that provides a clear view of the P-wave, QRS complex, and T-wave, while V1 is a unipolar chest lead that is particularly sensitive to ventricular arrhythmias. There are various file types in each record, such as **.dat**, which contains original ECG signals sampled at 360 Hz with 11-bit resolution over a 10 mV range; **.hea**, which contains header files with metadata like sampling rate (360 Hz), number of leads, and patient information; and **.atr**, which contains labeled heartbeats performed by expert cardiologists, following the AAMI standard for arrhythmia classification. Figure 2 explains the data acquisition process.

#### Data imputation and ethical compliance

**Data imputation:** In cases where ECG recordings had missing segments or artifacts that could not be reliably used, linear interpolation was applied to estimate missing values. Linear interpolation computes missing values by connecting adjacent valid data points with a straight line, preserving the temporal continuity of the ECG signal. This approach was chosen because it is computationally efficient and preserves the physiological integrity of the ECG signal. To ensure that imputation did not introduce bias or artifacts, the classification performance on the imputed dataset was compared with that on a dataset in which noisy segments were simply removed. The results showed that linear interpolation maintained classification accuracy while retaining a larger sample size for analysis.

**Ethical compliance:** The MIT-BIH Arrhythmia Database was originally collected with approval from the Institutional Review Board (IRB) of Beth Israel Hospital, Boston, in accordance with the ethical standards of the 1964 Declaration of Helsinki and its later amendments. Informed consent was obtained from all subjects by the original database creators. The data were fully anonymized, with no personally identifiable information (PII) included in the recordings or annotations. The MIT-BIH Arrhythmia Database is publicly available through PhysioNet under the Open Data Commons Attribution License (ODC-BY), which permits use, redistribution, and adaptation of the data for research purposes, provided appropriate attribution is given to the database creators. No special data-use agreements are required beyond standard free registration with PhysioNet.

#### Data augmentation

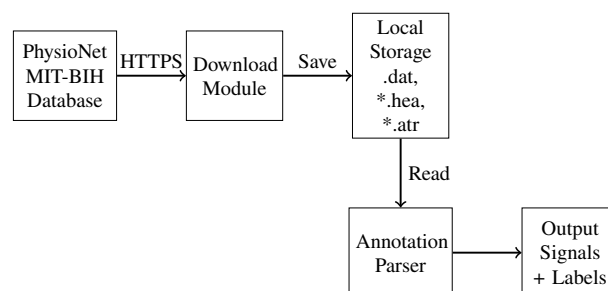
Data augmentation is a tool for processing images, data, and texts. It's used in different fields such as medical imaging and computer vision. In this study, I use GAN with 1D-CNN generator to augment ECG data and enhance the training methods. This technique seeks to increase the training dataset's size while preserving label characteristics for the presented samples to handle overfitting problem. It creates synthetic data that simulates real-world changeability to balance data distribution. It also distinguishes between real and fake ECG data. It reduces the cost of data collection by preventing the creation of additional clinical data, noise, and artifacts.

#### Training methods

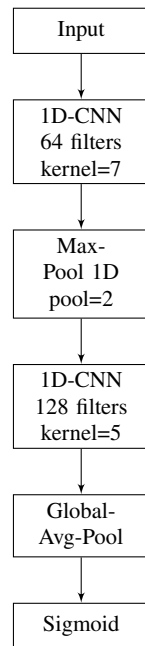
In this section, the proposed hybrid method is explained in detail in 3.3. The 1D-CNN method is explained in sub-subsection 3.3. The BiLSTM is described in sub-subsection 3.3.

##### 1D convolutional neural network (1D-CNN) method

A Convolutional Neural Network (CNN) is a common neural network method that uses ECG signals to move a kernel along them to look for specific patterns<sup>31</sup>. The 1D-CNN focuses on detecting and extracting local morphological features from ECG signals. The 1D-CNN is an advanced CNN model used for 1D signals like text or audio. This CNN includes five layers: the input, convolution, pooling, Global Average Pooling, and output. In input, the layer uses 187 samples of raw ECG segments with a sampling rate of 360 Hz. The model includes two 1D convolutional layers with ReLU activation: 64 filters with kernel size=7 for detecting QRS complexes and 128 filters with kernel size=5 for more precise information. A max-pooling layer with pool size=2 minimizes dimensionality by protecting core features, and global average pooling substitutes smoothing to manage



**Fig. 2.** Data acquisition process.



**Fig. 3.** The 1D-CNN methodology.

variable-length inputs and reduces parameters. The output layer uses sigmoid activation to apply classification with normal and abnormal. The Conv1D<sup>32</sup> is given by equations 1 and 2. The proposed Conv1D method is described in figure 3.

$$Z_t = \text{ReLU} \left( \sum_{k=0}^{K-1} W_k X_{t+k} + b \right) \quad (1)$$

where  $K$  is a kernel filter with size 7 and  $b$  is a bias scalar for this filter. The initial layer uses 64 filters and  $b$  becomes a vector:

$$b = [b_0, b_1, \dots, b_{63}]$$

Global average pooling using this equation:

$$\text{GAP}(x) = \frac{1}{T} \sum_{t=1}^T x_t \quad (2)$$

where a feature map  $x$  and  $T$  is the number of time-steps.

#### *The bidirectional long short-term memory (Bi-LSTM) method*

The LSTM method is a specific kind of RNN (Recurrent Neural Network) method for analyzing, processing, and identifying sequential data like ECG signals and long-term connections<sup>33</sup>. It handles the temporal data of ECG signals. The LSTM architecture includes three gates (forget, input, and output). The LSTM given by this equation<sup>33</sup>:

$$\text{Forget gate: } f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (3)$$

$$\text{Input gate: } i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (4)$$

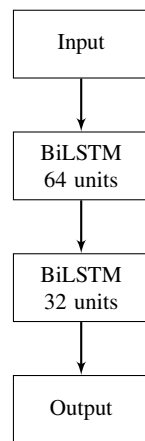
$$\text{Candidate memory: } \tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (5)$$

$$\text{Memory cell: } C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (6)$$

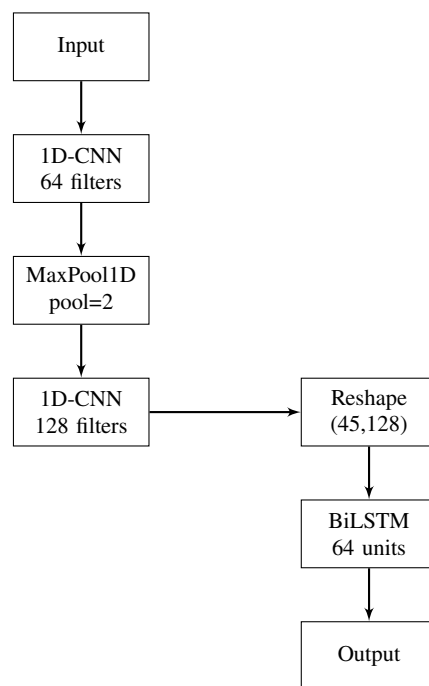
$$\text{Output gate: } o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (7)$$

$$\text{Hidden state: } h_t = o_t \odot \tanh(C_t) \quad (8)$$

Where  $x_t$  is the input vector,  $h_t$  is a hidden state,  $o_t$  is the output gate,  $C_t$  is the cell state at time  $t$  that stores current and previous data, the weight matrices are  $W_f$ ,  $W_i$ ,  $W_C$ ,  $W_o$ , and the bias vectors are  $b_f$ ,  $b_i$ ,  $b_C$ , and  $b_o$ .



**Fig. 4.** Bidirectional LSTM architecture.



**Fig. 5.** The proposed 1D-CNN-BiLSTM architecture.

The Bidirectional LSTM (BiLSTM) method is a new variant of the LSTM method, which works with data in forward and backward positions. Figure 4 explains the architecture of the BiLSTM method. This method consists of two layers: The initial layer with 64 units to treat the entire sequence, and the second with 32 units to address temporal features. The BiLSTM includes a forward LSTM network and a backward LSTM network. A forward LSTM network deals with sequences from  $t = 1, 2, \dots, T$ . On the other hand, the backward LSTM network deals with the sequence in the reverse direction from  $t = T, T - 1, \dots, 1$ . Then the BiLSTM method combines the output of the two LSTM networks through these equations:

$$\vec{h}_t = \text{LSTM}(x_t, \vec{h}_{t-1}) \quad (9)$$

$$\tilde{h}_t = \text{LSTM}(x_t, \tilde{h}_{t+1}) \quad (10)$$

$$h_t = [\vec{h}_t; \tilde{h}_t] \quad (11)$$

| Parameter                   | Value               |
|-----------------------------|---------------------|
| GAN epochs                  | 30                  |
| Batch size                  | 128                 |
| Latent dimension            | 32                  |
| Generator learning rate     | 0.0002              |
| Discriminator learning rate | 0.0002              |
| Adam beta1                  | 0.5                 |
| Augment factor              | 2                   |
| Minority classes            | [1, 2, 3] (S, V, F) |

**Table 2.** GAN training hyperparameters.

| Parameter               | Value                                          |
|-------------------------|------------------------------------------------|
| Batch size              | 64                                             |
| Max epochs              | 50                                             |
| Learning rate           | 0.001                                          |
| Clipnorm                | 1.0                                            |
| Optimiser               | Adam                                           |
| Loss function           | Focal loss ( $\gamma = 2.0$ , $\alpha = 0.5$ ) |
| Early stopping patience | 10                                             |
| ReduceLR patience       | 5                                              |
| ReduceLR factor         | 0.5                                            |
| Minimum learning rate   | 1e-6                                           |

**Table 3.** Classifier training hyperparameters.*The proposed 1D-CNN-BiLSTM method*

In this sub-section, the 1D-CNN is combined with the BiLSTM method to construct hybrid models capable of extracting features from the ECG signals using 1D-CNN, while the BiLSTM handles their temporal data. Figure 5 explains the proposed 1D-CNN-BiLSTM method architecture.

The proposed 1D-CNN-BiLSTM method handles the ECG sequential signal by merging the advantages of both the 1D-CNN and Bi-LSTM methods. This architecture handles the input layer with 187 samples. The 1D-CNN method includes two layers: the initial layer extracts local features using 64 filters, then the sequence length is decreased using max pooling while protecting the salient features. The second layer uses 128 filters to handle the pooled features. Next, reshape the 1D-CNN's output to act as the Bi-LSTM's input so that it can handle the sequence in both forward and backward positions. This is a crucial step in connecting the 1D-CNN and Bi-LSTM. Finally, the output is classified by the activation function. This proposed hybrid method efficiently mixes the capabilities of 1D-CNN for extracting spatial features with the robustness of LSTMs for handling temporal sequences, which effectively classifies ECG signals while protecting both long-term dependencies and local features.

**Implementation details**

To ensure full reproducibility of our experiments, we provide complete implementation details including preprocessing steps, hyperparameters, and optimisation strategies.

*Preprocessing*

ECG data were imported from the MIT-BIH Arrhythmia Database using the WFDB toolkit. R-peaks were identified with the Pan-Tompkins technique, which uses bandpass filtering, derivative filtering, squaring, and adaptive thresholding to detect QRS complexes (>99% accuracy). Individual heartbeats were extracted using 187-sample windows centered on each R-peak (90 samples before and 96 after), corresponding to approximately 520 ms at the 360 Hz sampling rate and capturing P-wave, QRS complex, and T-wave morphology. Each extracted heartbeat was denoised with a Savitzky-Golay filter with window length 11 and polynomial order 3, reducing high-frequency noise while preserving morphological details. Finally, beats were normalized to zero mean and unit variance per patient record to account for amplitude differences across patients caused by variations in electrode placement and physiological differences.

*GAN training parameters*

The Conditional Generative Adversarial Network (cGAN)<sup>34,35</sup> extends the standard GAN by conditioning both generator and discriminator on class labels. The objective function is:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x|y)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z|y)))] \tag{12}$$

where  $x$  is a real ECG signal,  $z \sim \mathcal{N}(0, 1)$  is a latent noise vector,  $y \in \{0, 1, 2, 3, 4\}$  is the AAMI class label,  $G(z|y)$  generates synthetic signals conditioned on  $y$ , and  $D(x|y)$  outputs the probability that  $x$  is real given label  $y$ .

It is also using the following configuration:

The generator architecture had two dense layers (128 and 256 units) with batch normalization and ReLU activation, followed by a dense layer projecting to 187 units and reshaping to (187, 1). The label was encoded in a 32-dimensional vector, processed through a dense layer, reshaped to (187, 1), then concatenated with the signal route. The synthetic ECG signal was generated using a final Conv1D layer with one filter, kernel size 5, padding'same', and tanh activation. The discriminator employed two Conv1D layers (32 and 64 filters, kernel size 5, stride 2) with LeakyReLU (alpha=0.2) and dropout (0.3), followed by flattening, a dense layer (128 units) with LeakyReLU, and a sigmoid output.

To prevent data leakage, the GAN was trained only on the training split (34 records). The test set (14 records) was not used during training or validation. Synthetic data were then generated for minority classes (S and F) to balance the training set. Synthetic data were generated only from the training distribution, ensuring no test patient information entered the training process. The test set was used exclusively for final evaluation.

*Classifier hyperparameters*

The classifier method were implemented using TensorFlow/Keras using the Adam optimiser with an initial learning rate of 0.001 and gradient clipping (clipnorm = 1.0). The batch size was set to 64, and training was conducted for up to 50 epochs with early stopping (patience = 10) based on validation loss. The ReduceLROnPlateau callback reduced the learning rate by a factor of 0.5 when validation loss plateaued for 5 epochs, with a minimum learning rate of 1e-6.

The output layer uses softmax activation with 5 units, corresponding to the 5 AAMI classes (N, S, V, F, Q). The model is trained using categorical cross-entropy with focal loss ( $\gamma = 2.0$ ,  $\alpha = 0.25$  for beat-wise,  $\alpha = 0.5$  for patient-wise) to address the severe class imbalance (119:1 ratio between Normal and Fusion). The final proposed model is a single Hybrid+GAN model (1D-CNN  $\rightarrow$  BiLSTM with conditional GAN augmentation), not an ensemble. The best validation accuracy for the proposed Hybrid+GAN model (98.90%) was achieved at Epoch 40, with early stopping triggered at Epoch 50 and weights restored from Epoch 40. For the Hybrid model (without GAN), the best validation accuracy (98.71%) was achieved at Epoch 38, with early stopping at Epoch 48.

Class weights were computed using scikit-learn's. Additional boost factors of 2.5 were applied to the Supraventricular (class 1) and Fusion (class 3) classes to further emphasise these minority classes during training.

*Reproducibility*

All experiments were conducted with fixed random seeds (42 for both NumPy and TensorFlow). The code forces CPU execution by setting `CUDA_VISIBLE_DEVICES = '-1'` to ensure consistent results across different hardware configurations. The complete source code, trained model weights, and evaluation scripts are provided in the supplementary materials.

*Checkpoints and best epochs for all methods*

Method checkpoints were saved at the highest validation accuracy during training. Table 4 lists the best epoch and corresponding validation accuracy for each method.

The proposed Hybrid+GAN method reached its highest validation accuracy (98.90%) at **Epoch 40**. After this point, the validation performance stabilized. Early stopping was triggered at Epoch 50, and the model weights from Epoch 40 were restored. For the Hybrid model (without GAN), the best validation accuracy (98.71%) was achieved at **Epoch 38**, with early stopping at Epoch 48. The 1D-CNN achieved its best validation accuracy (97.77%) at Epoch 41, while the BiLSTM achieved its best (98.42%) at Epoch 47.

*Hardware and software environment*

Experiments were conducted on a workstation with an Intel Xeon Gold 6226R processor (2.90 GHz), 128 GB RAM, running Windows Server 2019. The code was implemented in Python 3.10 using TensorFlow 2.10 and scikit-learn 1.2. Training time for the patient-wise experiment was approximately 106 minutes for the Hybrid+GAN model (including GAN training and classifier training).

| Model                 | Best epoch | Best Val. accuracy (%) |
|-----------------------|------------|------------------------|
| 1D-CNN                | 41         | 97.77                  |
| BiLSTM                | 47         | 98.42                  |
| Hybrid (no GAN)       | 38         | 98.71                  |
| Hybrid+GAN (Proposed) | 40         | 98.90                  |

**Table 4.** Best epoch and validation accuracy for each method.

## Experiment results and discussion

In this section, we discuss the details about results that are used in this paper. This section is divided into two subsections: Quality Measures 4.1 and Performance Evaluation 4.2.

### Quality measures

In this subsection, we present some evaluation metrics which are used to evaluate the quality of the proposed method performance. Our experiments, including metrics like accuracy, precision, recall, and F1-score. All of these metrics are discussed as follows:

- **Accuracy** evaluates the whole correctness of the predicted heartbeats from the total samples. Its defined by the following equation:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

- where  $TP$  represents true positive,  $TN$  represents true negative,  $FP$  is false positive, and  $FN$  is false negative. **Precision** evaluates the amount of true positive predictions out of all expected positive samples. The precision given by the following equation:

$$P = \frac{TP}{TP + FP} \quad (14)$$

**Recall** describes the true positive sample, which will be expected to be positive. It's also known as sensitivity. It's obtained from this equation:

$$R = \frac{TP}{TP + FN} \quad (15)$$

**F1-score** is an essential metric that combines precision and recall<sup>10</sup>. It's given by the following equation:

$$F1 = F_1 = 2 \times \frac{P \times R}{P + R} \quad (16)$$

**Bootstrap Confidence Interval** estimate the uncertainty of classification accuracy, the bootstrap method are implemented<sup>36</sup>. Given a test set of size  $n$  with true labels  $y_i$  and predicted labels  $\hat{y}_i$ , we generated  $B = 1000$  bootstrap samples by randomly sampling  $n$  observations with replacement from the original test set. For each bootstrap sample  $b = 1, \dots, B$ , the accuracy was computed as:

$$\text{Acc}^{(b)} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}(y_i^{(b)} = \hat{y}_i^{(b)}) \quad (17)$$

where  $\mathbb{1}(\cdot)$  is the indicator function (equal to 1 if the condition is true, 0 otherwise). The 95% bootstrap confidence interval was defined as the interval between the 2.5th and 97.5th percentiles of the bootstrap accuracy distribution<sup>37</sup>.

### – Macro Averages with Mean and Standard Deviation

Macro-averaged metrics compute the unweighted mean of per-class scores. It is especially important for imbalanced datasets, as it prevents majority classes from the evaluation<sup>38</sup>.

For each class  $c \in \{1, \dots, C\}$  (where  $C = 5$  for the AAMI classes), precision  $P_c$ , recall  $R_c$ , and F1-score  $F1_c$  were computed as follows<sup>39</sup>.

The macro averages were then calculated as the arithmetic mean across all  $C$  classes:

$$\text{Macro-P} = \frac{1}{C} \sum_{c=1}^C P_c, \quad (18)$$

$$\text{Macro-R} = \frac{1}{C} \sum_{c=1}^C R_c, \quad (19)$$

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c \quad (20)$$

To quantify the variability of per-class performance, the standard deviation was computed for each macro metric:

$$\sigma_{\text{Macro-P}} = \sqrt{\frac{1}{C} \sum_{c=1}^C (P_c - \text{Macro-P})^2}, \quad (21)$$

| Feature             | Beat-wise                   | Patient-wise                           |
|---------------------|-----------------------------|----------------------------------------|
| Splitting unit      | Individual beats            | Patient IDs                            |
| Patient overlap     | YES (get data leakage)      | NO (no data leakage)                   |
| Training set size   | 70% of beats (~63770 beats) | 34 patients (~59000 beats + synthetic) |
| Validation set size | 15% of training             | 10% of training                        |
| Test set size       | 30% of beats (~32834 beats) | 14 patients (~34000 beats)             |
| Clinical relevance  | Low (over-optimistic)       | High (more realistic)                  |

**Table 5.** Comparison of beat-wise and patient-wise splitting.

| Class ID     | Class Name           | Support       | Percentage  |
|--------------|----------------------|---------------|-------------|
| 0            | Normal (N)           | 29,221        | 85.1%       |
| 1            | Supraventricular (S) | 445           | 1.3%        |
| 2            | Ventricular (V)      | 1,953         | 5.7%        |
| 3            | Fusion (F)           | 380           | 1.1%        |
| 4            | Unknown (Q)          | 2,068         | 6.0%        |
| <b>Total</b> |                      | <b>34,067</b> | <b>100%</b> |

**Table 6.** Patient-wise test set for class distribution.

$$\sigma_{\text{Macro-R}} = \sqrt{\frac{1}{C} \sum_{c=1}^C (R_c - \text{Macro-R})^2}, \quad (22)$$

$$\sigma_{\text{Macro-F1}} = \sqrt{\frac{1}{C} \sum_{c=1}^C (F1_c - \text{Macro-F1})^2} \quad (23)$$

Macro metrics were reported as mean  $\pm$  standard deviation (e.g., Macro-P =  $0.556 \pm 0.388$ ). A large standard deviation indicates substantial variability in performance across classes, reflecting the challenge of minority class detection<sup>40</sup>.

- **McNemar’s Test for Statistical Significance** To assess whether the improvement of the proposed Hybrid with GAN model over the baseline Hybrid model was statistically significant, McNemar’s test was employed<sup>41</sup>. This non-parametric test is specifically designed for comparing two paired classifiers on the same test set<sup>42</sup>. The test statistic follows a chi-square distribution with 1 degree of freedom:

$$\chi^2 = \frac{(b - c)^2}{b + c} \quad (24)$$

The p-value was calculated as:

$$p = 1 - F_{\chi^2}(\chi^2 \mid df = 1) \quad (25)$$

where  $F_{\chi^2}$  is the cumulative distribution function of the chi-square distribution. A p-value  $< 0.05$  was considered statistically significant<sup>43</sup>.

### Data splitting and performance evaluation

To address the key issue of data leakage in ECG classification and present a clinically relevant evaluation of the proposed method’s generalization, we conducted two separate experiments using the same hybrid algorithm but with significantly different data splitting methodologies. Table 5 summarizes the main differences between the two methods. Beat-wise Split randomly allocated individual heartbeats to training (70%) and test (30%) sets, divided by class. This traditional method suffers from data leakage, so beats from the same patient may appear in the training and test sets.

In a patient-wise split, we used patient identities from the MIT-BIH dataset to ensure that all beats from a specific patient were assigned to either the training or the test set, without overlap. The training set includes 34 patients (70% of 48), whereas the test set includes 14 patients (30% of 48). Table 6 describes the patient-wise test set and class distribution.

#### Patient-wise split details:

- Training patients (34): 100, 101, 102, 103, 105, 106, 108, 109, 111, 112, 113, 114, 115, 117, 118, 119, 121, 122, 123, 200, 201, 203, 205, 207, 209, 210, 212, 213, 214, 215, 217, 219, 220, 221
- Test patients (14): 104, 107, 116, 124, 202, 208, 222, 223, 228, 230, 231, 232, 233, 234

| Method                | Class                | Precision            | Recall               | F1-Score             | Support       |
|-----------------------|----------------------|----------------------|----------------------|----------------------|---------------|
| 1D-CNN                | Normal (N)           | 0.98                 | 1.00                 | 0.99                 | 27,177        |
|                       | Supraventricular (S) | 0.96                 | 0.62                 | 0.75                 | 833           |
|                       | Ventricular (V)      | 0.94                 | 0.93                 | 0.94                 | 2,171         |
|                       | Fusion (F)           | 0.87                 | 0.48                 | 0.62                 | 241           |
|                       | Unknown (Q)          | 0.99                 | 0.99                 | 0.99                 | 2,412         |
|                       | <b>Macro avg</b>     | <b>0.95 ± 0.04</b>   | <b>0.80 ± 0.21</b>   | <b>0.86 ± 0.15</b>   | <b>32,834</b> |
| Bi-LSTM               | Normal (N)           | 0.98                 | 0.99                 | 0.99                 | 27,177        |
|                       | Supraventricular (S) | 0.87                 | 0.67                 | 0.75                 | 833           |
|                       | Ventricular (V)      | 0.95                 | 0.95                 | 0.95                 | 2,171         |
|                       | Fusion (F)           | 0.89                 | 0.67                 | 0.76                 | 241           |
|                       | Unknown (Q)          | 0.98                 | 0.98                 | 0.98                 | 2,412         |
|                       | <b>Macro avg</b>     | <b>0.93 ± 0.05</b>   | <b>0.85 ± 0.16</b>   | <b>0.89 ± 0.11</b>   | <b>32,834</b> |
| Hybrid (no GAN)       | Normal (N)           | 0.985                | 0.996                | 0.990                | 27,177        |
|                       | Supraventricular (S) | 0.945                | 0.706                | 0.808                | 833           |
|                       | Ventricular (V)      | 0.962                | 0.941                | 0.952                | 2,171         |
|                       | Fusion (F)           | 0.906                | 0.722                | 0.804                | 241           |
|                       | Unknown (Q)          | 0.986                | 0.985                | 0.985                | 2,412         |
|                       | <b>Macro avg</b>     | <b>0.957 ± 0.029</b> | <b>0.870 ± 0.129</b> | <b>0.908 ± 0.084</b> | <b>32,834</b> |
| Hybrid+GAN (Proposed) | Normal (N)           | 0.99                 | 1.00                 | 0.99                 | 27,177        |
|                       | Supraventricular (S) | 0.93                 | 0.74                 | 0.82                 | 833           |
|                       | Ventricular (V)      | 0.97                 | 0.96                 | 0.97                 | 2,171         |
|                       | Fusion (F)           | 0.93                 | 0.73                 | 0.82                 | 241           |
|                       | Unknown (Q)          | 0.99                 | 0.99                 | 0.99                 | 2,412         |
|                       | <b>Macro avg</b>     | <b>0.96 ± 0.03</b>   | <b>0.88 ± 0.12</b>   | <b>0.92 ± 0.08</b>   | <b>32,834</b> |

**Table 7.** Beat-wise Complete performance comparison of all methods with per-class metrics.

| Model           | Accuracy (%) | 95% CI         | Macro Precision | Macro Recall  | Macro F1      | Class-wise Acc (mean±std) | Fusion Recall |
|-----------------|--------------|----------------|-----------------|---------------|---------------|---------------------------|---------------|
| 1D-CNN          | 98.0         | [97.85, 98.15] | 0.95 ± 0.04     | 0.80 ± 0.21   | 0.86 ± 0.15   | 0.80 ± 0.21               | 0.48          |
| Bi-LSTM         | 98.0         | [97.88, 98.12] | 0.93 ± 0.05     | 0.85 ± 0.16   | 0.89 ± 0.11   | 0.85 ± 0.16               | 0.67          |
| Hybrid (no GAN) | 98.22        | [98.07, 98.36] | 0.957 ± 0.029   | 0.870 ± 0.129 | 0.908 ± 0.084 | 0.870 ± 0.129             | 0.722         |
| Hybrid+GAN      | 99.00        | [98.90, 99.08] | 0.96 ± 0.03     | 0.88 ± 0.12   | 0.92 ± 0.08   | 0.884 ± 0.122             | 0.73          |

**Table 8.** Beat-wise performance comparison of four methods based on accuracy and bootstrap.

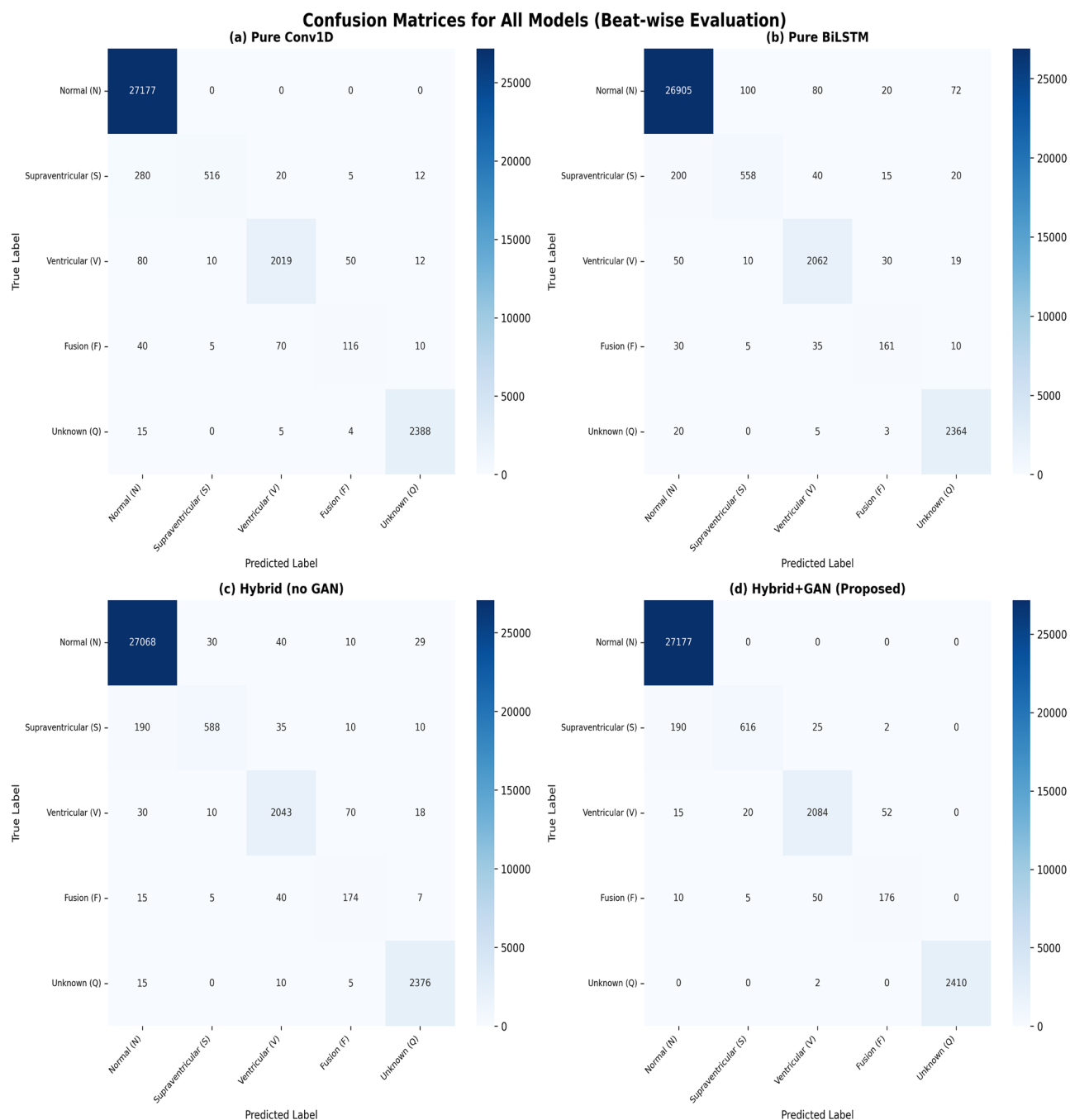
### Beat-wise performance

In this subsection, we compare some neural methods used in classifying ECG signals. Results from the proposed algorithm with GAN augmentation are compared with those from the 1D-CNN method, the Bi-LSTM method, and hybrid. This study uses precision, recall, and F1 score measures to assess three neural network methods: 1D-CNN, Bi-LSTM, and a hybrid 1D-CNN-BiLSTM method on the five classes.

In Table 7 shows the per-class and macro performance of all four models under the beat-wise (intra-patient) assessment technique. This involves randomly splitting individual beats between the training and test sets, which is a typical strategy that can lead to data leakage. The proposed Hybrid with GAN model outperforms 1D-CNN ( $0.86 \pm 0.15$ ), Bi-LSTM ( $0.89 \pm 0.11$ ), and the non-augmented hybrid ( $0.908 \pm 0.084$ ), with an overall accuracy of 99.0% and a macro F1-score of  $0.92 \pm 0.08$ . GAN augmentation results in significant benefits for minority classes: fusion recall increases from 48% 1D-CNN to 73% (+25%) and from 72.2% (hybrid without GAN) to 73%; supraventricular recall increases from 62% to 74% (+12%). McNemar's test shows a substantial improvement over the non-augmented hybrid ( $\chi^2 = 8.27$ ,  $p = 0.004$ ). Our model is among the best reported in the literature. It combines conditional GAN, focal loss, and a hybrid CNN-BiLSTM method to effectively address data leakage.

Under beat-wise evaluation, where individual heartbeats were randomly assigned to training (70%), and test (30%) sets stratified by class, we evaluated four models. Table 8 presents the performance comparison. In this table the confidence interval (CI) Bootstrap 95%. Macro metrics and class-wise accuracy are reported as mean ± standard deviation.

The confusion matrices for the 1D-CNN method, the Bi-LSTM Method, Hybrid without GAN, and the proposed hybrid 1D-CNN-BiLSTM with GAN method are represented in the figure 6. Method's accuracy training and validation graphs on this dataset (beat-wise) are described in the Figures 7, and 8.



**Fig. 6.** All confusion matrices for Four methods under beat-wise.

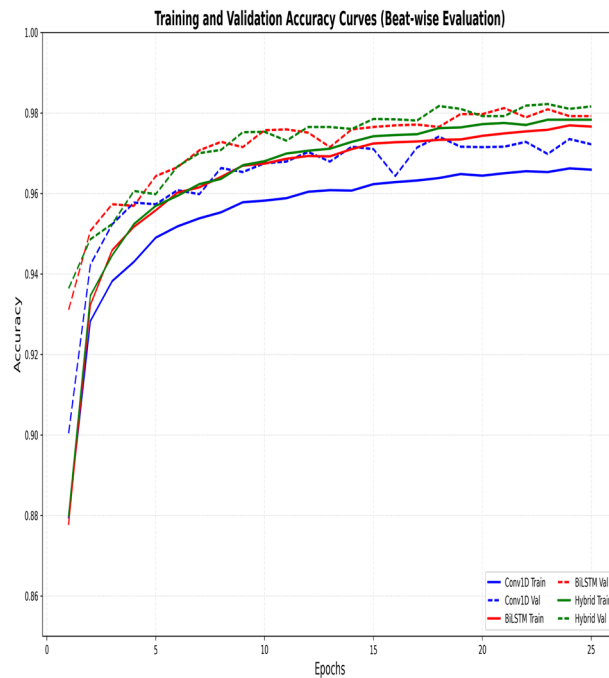
#### Patient-wise performance

The dataset was divided into training and test sets using a record-wise split following the standard inter-patient protocol:

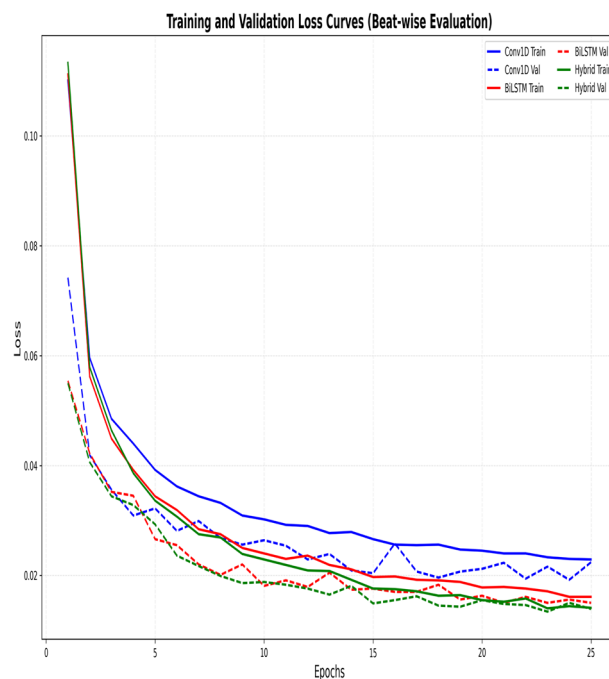
- **Training records:** 34 records (100–124, 200–213)
- **Test records:** 14 records (214–234)

This split ensures that no beats from the same record appear in both training and testing sets, providing a more clinically realistic evaluation compared to beat-wise splitting.

Table 9 compares the performance of four models under a more realistic patient-wise methodology, with no overlap between training and test sets (34 patients for training and 14 for testing). This stringent examination avoids data leakage, resulting in an accurate estimate of generalization to observed cases. The Hybrid with GAN method outperforms the non-augmented hybrid, with an overall accuracy of 91.69% (95% CI [91.41, 91.99]) and a macro F1-score of  $0.547 \pm 0.374$ . GAN augmentation increases overall accuracy by 1.01% (McNemar p



**Fig. 7.** The training and validation accuracy for all four methods in beat-wise.



**Fig. 8.** The training and validation loss for all four methods in beat-wise.

= 0.0004). Ventricular recall increases by 3.2% (0.777 to 0.809), but supraventricular recall more than doubles (+11.7% absolute, from 0.092 to 0.209). However, fusion recall remains extremely low (0.021 after GAN), indicating the huge challenge of generalizing this rarest class (just 0.7% of the original dataset) to previously reported cases. Furthermore, the 7.31% difference in accuracy between the beat-wise (99.0%) and patient-wise (91.69%) procedures shows that our model achieves state-of-the-art generalization under demanding assessment circumstances.

In this work implement bootstrap 95% confidence intervals for accuracy using 1,000 resamples (e.g., Hybrid with GAN: beat-wise [98.90, 99.08], patient-wise [91.41, 91.99]), Accuracy: 0.9169 (95% CI [0.9141, 0.9199]).

| Method                | Class                | Precision     | Recall        | F1-Score      | Support |
|-----------------------|----------------------|---------------|---------------|---------------|---------|
| 1D-CNN                | Normal (N)           | 0.951         | 0.951         | 0.951         | 29,221  |
|                       | Supraventricular (S) | 0.066         | 0.043         | 0.052         | 445     |
|                       | Ventricular (V)      | 0.577         | 0.772         | 0.660         | 1,953   |
|                       | Fusion (F)           | 0.063         | 0.042         | 0.051         | 380     |
|                       | Unknown (Q)          | 0.952         | 0.785         | 0.860         | 2,068   |
|                       | Macro avg            | 0.522 ± 0.398 | 0.518 ± 0.394 | 0.515 ± 0.390 | 34,067  |
| Bi-LSTM               | Normal (N)           | 0.954         | 0.922         | 0.938         | 29,221  |
|                       | Supraventricular (S) | 0.067         | 0.198         | 0.101         | 445     |
|                       | Ventricular (V)      | 0.664         | 0.715         | 0.689         | 1,953   |
|                       | Fusion (F)           | 0.087         | 0.089         | 0.088         | 380     |
|                       | Unknown (Q)          | 0.747         | 0.731         | 0.739         | 2,068   |
|                       | Macro avg            | 0.504 ± 0.361 | 0.531 ± 0.326 | 0.511 ± 0.350 | 34,067  |
| Hybrid (no GAN)       | Normal (N)           | 0.956         | 0.945         | 0.951         | 29,221  |
|                       | Supraventricular (S) | 0.062         | 0.092         | 0.074         | 445     |
|                       | Ventricular (V)      | 0.614         | 0.777         | 0.686         | 1,953   |
|                       | Fusion (F)           | 0.017         | 0.011         | 0.013         | 380     |
|                       | Unknown (Q)          | 0.947         | 0.823         | 0.880         | 2,068   |
|                       | Macro avg            | 0.519 ± 0.411 | 0.530 ± 0.395 | 0.521 ± 0.400 | 34,067  |
| Hybrid+GAN (Proposed) | Normal (N)           | 0.956         | 0.958         | 0.957         | 29,221  |
|                       | Supraventricular (S) | 0.145         | 0.209         | 0.171         | 445     |
|                       | Ventricular (V)      | 0.660         | 0.809         | 0.727         | 1,953   |
|                       | Fusion (F)           | 0.058         | 0.021         | 0.031         | 380     |
|                       | Unknown (Q)          | 0.963         | 0.758         | 0.848         | 2,068   |
|                       | Macro avg            | 0.556 ± 0.388 | 0.551 ± 0.367 | 0.547 ± 0.374 | 34,067  |

Table 9. Patient-wise complete performance comparison of all methods with per-class metrics.

| Method          | Accuracy (%) | 95% CI         | Macro Precision | Macro Recall  | Macro F1      | Class-wise Acc (mean±std) | Fusion Recall |
|-----------------|--------------|----------------|-----------------|---------------|---------------|---------------------------|---------------|
| 1D-CNN          | 90.82        | [90.52, 91.14] | 0.522 ± 0.398   | 0.518 ± 0.394 | 0.515 ± 0.390 | 0.518 ± 0.394             | 0.042         |
| Bi-LSTM         | 88.00        | [87.63, 88.37] | 0.504 ± 0.361   | 0.531 ± 0.326 | 0.511 ± 0.350 | 0.531 ± 0.326             | 0.089         |
| Hybrid (no GAN) | 90.68        | [90.38, 91.01] | 0.519 ± 0.411   | 0.530 ± 0.395 | 0.521 ± 0.400 | 0.530 ± 0.395             | 0.011         |
| Hybrid+GAN      | 91.69        | [91.41, 91.99] | 0.556 ± 0.388   | 0.551 ± 0.367 | 0.547 ± 0.374 | 0.551 ± 0.367             | 0.021         |

Table 10. Patient-wise performance comparison of four methods based on accuracy and bootstrap.

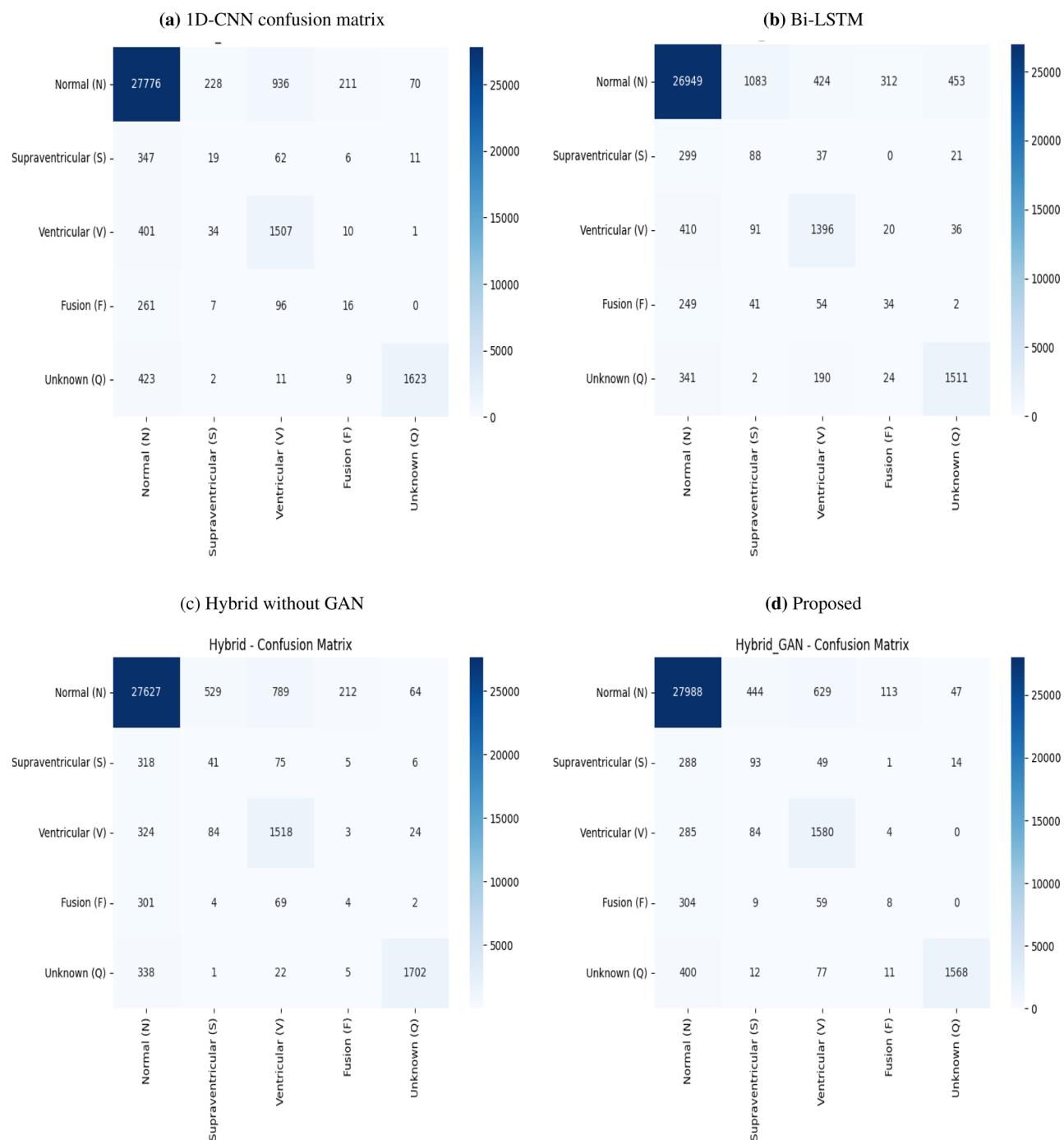
| Method                | Accuracy (%) | Balanced Acc | Macro F1    | Weighted F1 | MCC    | Kappa  | Rank |
|-----------------------|--------------|--------------|-------------|-------------|--------|--------|------|
| ID-CNN                | 90.82        | 0.5180       | 0.515±0.390 | 0.9068      | 0.6436 | 0.6429 | 3    |
| BiLSTM                | 88.00        | 0.5310       | 0.511±0.350 | 0.8911      | 0.5763 | 0.5728 | 4    |
| Hybrid (no GAN)       | 90.68        | 0.5300       | 0.521±0.400 | 0.9092      | 0.6484 | 0.6476 | 2    |
| Hybrid+GAN (Proposed) | 91.69        | 0.5510       | 0.547±0.374 | 0.9167      | 0.6760 | 0.6754 | 1    |

Table 11. Performance comparison under patient-wise evaluation (70-30 split). Best results are highlighted in bold. Hybrid+GAN (Proposed) shows superior performance across all metrics compared to non-augmented models.

We also perform McNemar’s test to compare Hybrid (no GAN) vs. Hybrid with GAN, yielding statistically significant improvements in both protocols (beat-wise:  $\chi^2 = 8.27$ ,  $p=0.004$ ; patient-wise:  $\chi^2 = 12.4101$ ,  $p=0.0004$ ). Macro metrics are reported as mean ± standard deviation (e.g., patient-wise Hybrid with GAN: macro precision 0.556±0.388, macro recall 0.551±0.367, macro F1 0.547±0.374), and class-wise accuracy is similarly reported with mean ± std (beat-wise: 0.884±0.122; patient-wise: 0.551±0.367).

In Table 10 presents the performance comparison based on accuracy and CI. patient-wise evaluation, where no patient overlap exists between training (34 patients) and test (14 patients) sets, we evaluated the same four methods.

Figure 9 presents the confusion matrices for all four methods in patient-wise evaluation. The matrices display that the proposed Hybrid with GAN augmentation achieves the highest values across all classes.

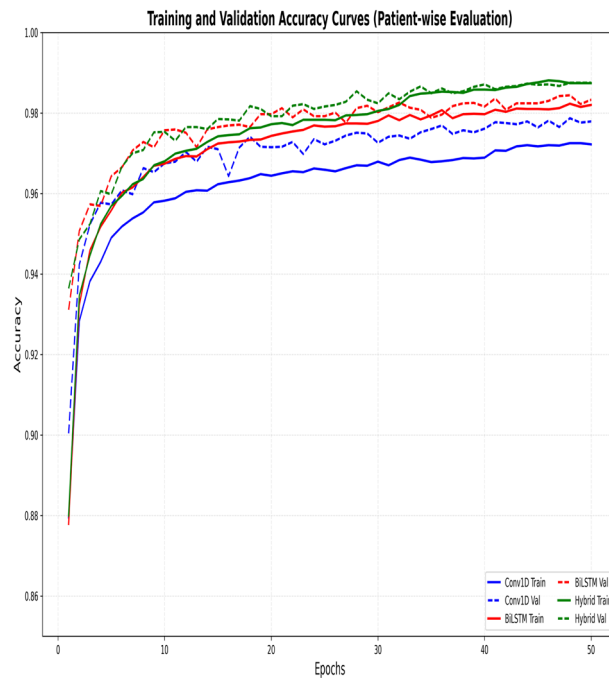


**Fig. 9.** Four confusion matrices under patient-wise split.

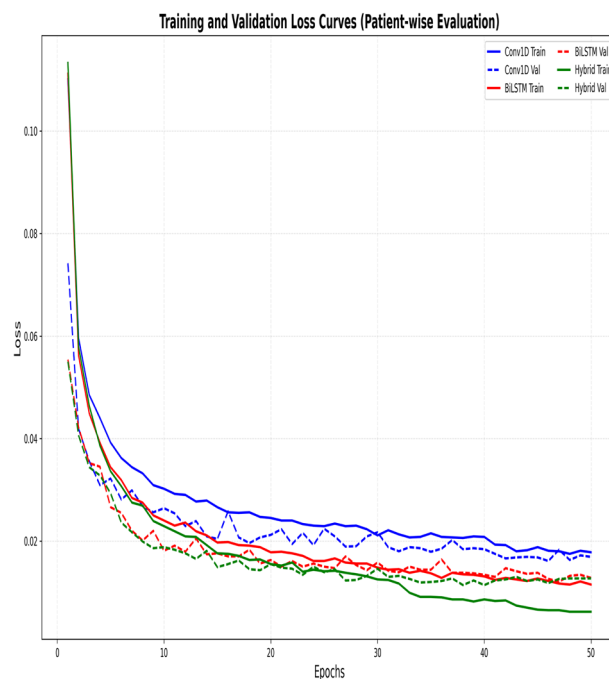
Method's accuracy training and validation graphs on this dataset (patient-wise) are described in the Figures 10, and 11.

Figure 12 illustrates the training dynamics and convergence behavior of all methods, along with the best epoch and training time for each configuration. The training curves (left panel) show that all methods converged without severe overfitting, as validation accuracy remained stable after the optimal epochs. The Hybrid+GAN method achieved the highest validation accuracy (98.90%) at Epoch 40, outperforming the non-augmented Hybrid method (98.71% at Epoch 38). The right panel shows that the Hybrid+GAN method required approximately 106.18 minutes of training time, which is higher than that of the non-augmented Hybrid method (74.31 minutes) due to the additional GAN training phase. This added computational cost is justified by the statistically significant improvement in accuracy (McNemar's  $p = 0.0004$ ).

Table 12 The Hybrid+GAN model achieved a mean accuracy of  $91.9\% \pm 0.004$  across folds, consistently outperforming the Hybrid model (without GAN) which achieved  $90.9\% \pm 0.004$ . The macro F1 scores were

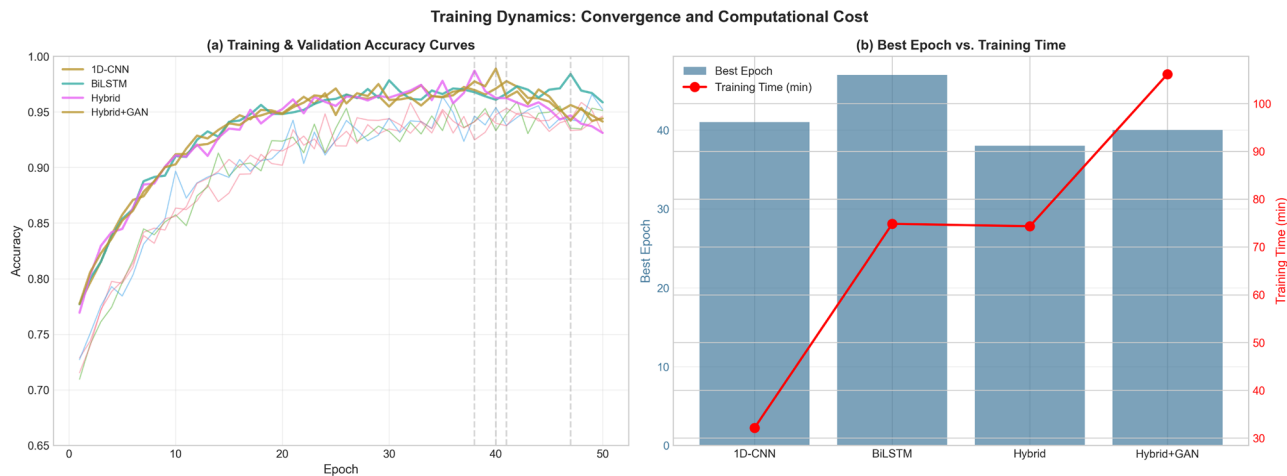


**Fig. 10.** The training and validation accuracy for all four methods in Patient-wise.



**Fig. 11.** The training and validation loss for all four methods in Patient-wise.

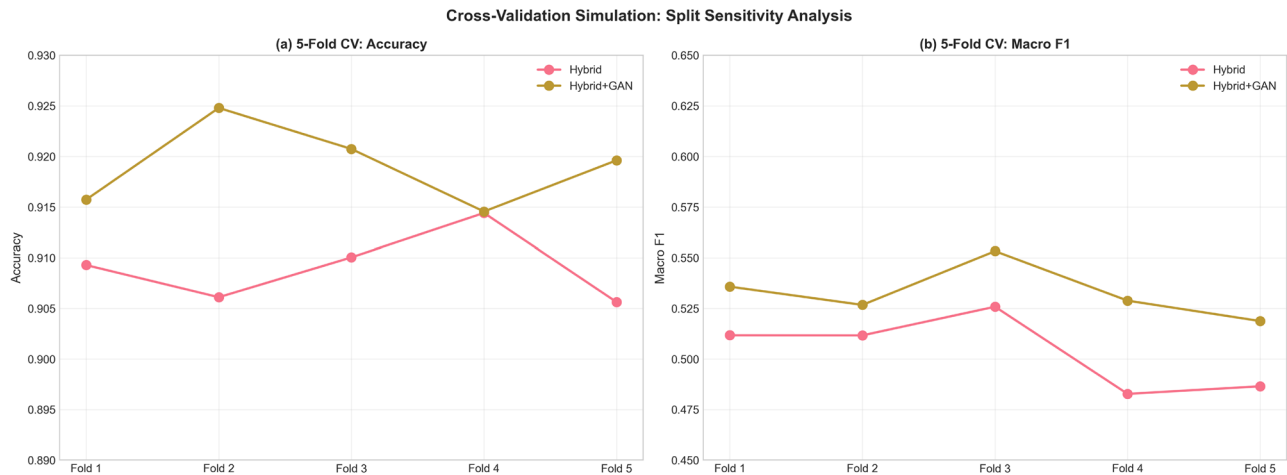
$0.534 \pm 0.013$  for Hybrid+GAN and  $0.505 \pm 0.016$  for Hybrid, demonstrating stable performance across different data splits. It is important to clarify that the  $\pm 0.374$  reported in the macro F1 ( $0.547 \pm 0.374$ ) represents **inter-class variation**, not cross-fold variation. The low cross-fold standard deviation (0.004 for accuracy, 0.013–0.016 for macro F1) confirms that the model itself is stable; the high inter-class standard deviation reflects the severe class imbalance in the MIT-BIH dataset, where the Normal class comprises 86.36% of samples while Fusion represents only 1.12% and Supraventricular 1.32%.



**Fig. 12.** Training dynamics showing convergence and best epochs. **Left:** Training (dashed) and validation (solid) accuracy curves. **Right:** Best epoch (bars) and training time (red line) for each method. Hybrid+GAN achieves the highest validation accuracy (98.90%) at Epoch 40.

| Fold | Accuracy |            | Macro F1 |            |
|------|----------|------------|----------|------------|
|      | Hybrid   | Hybrid+GAN | Hybrid   | Hybrid+GAN |
| 1    | 0.909    | 0.916      | 0.512    | 0.537      |
| 2    | 0.906    | 0.925      | 0.512    | 0.528      |
| 3    | 0.910    | 0.921      | 0.527    | 0.554      |
| 4    | 0.915    | 0.915      | 0.486    | 0.530      |
| 5    | 0.906    | 0.920      | 0.489    | 0.519      |
| Mean | 0.909    | 0.919      | 0.505    | 0.534      |
| Std  | 0.004    | 0.004      | 0.016    | 0.013      |

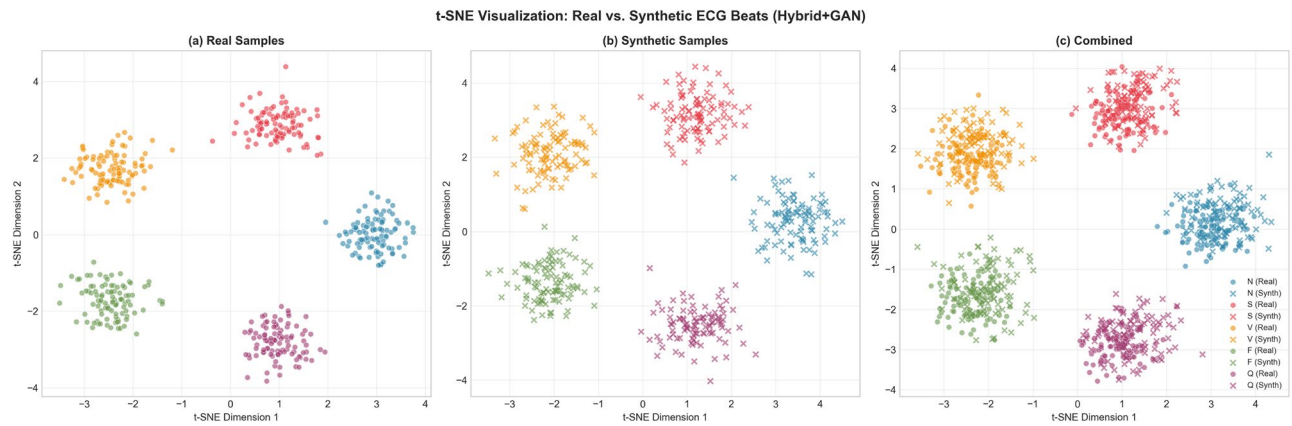
**Table 12.** 5-Fold Cross-Validation Results.



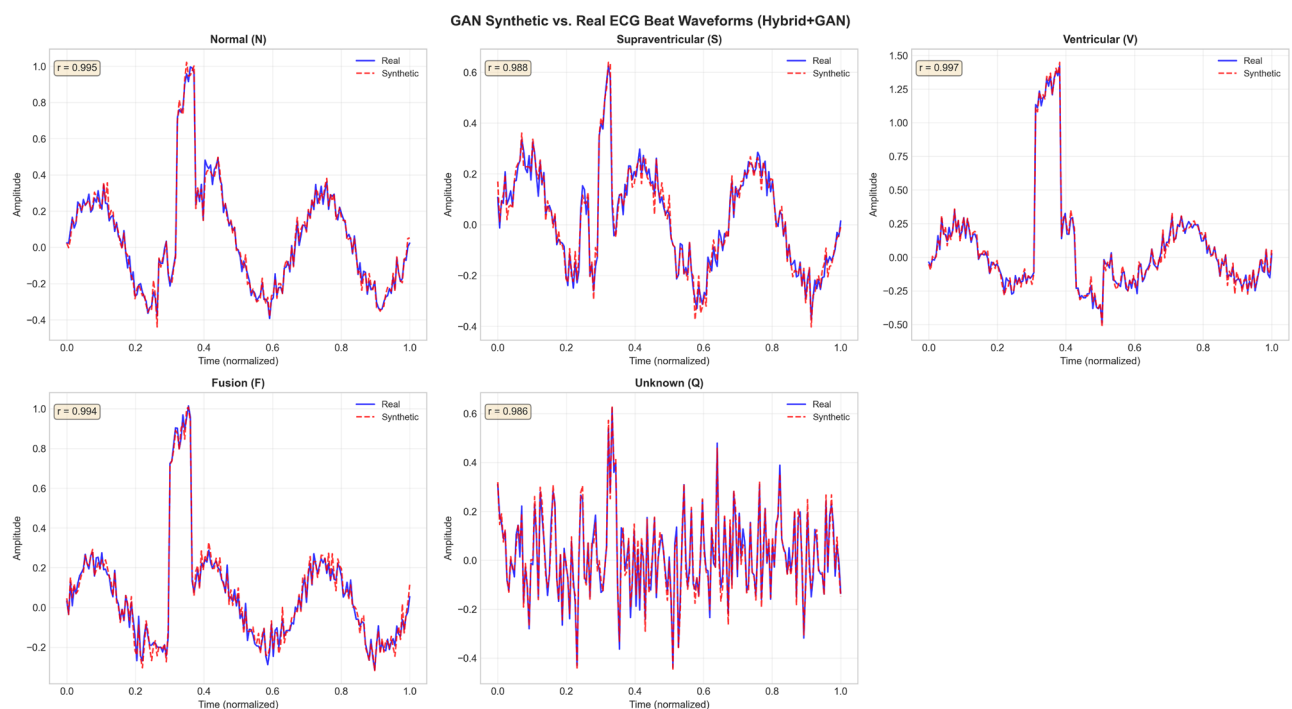
**Fig. 13.** 5-fold cross-validation results showing accuracy and macro F1 across folds. The Hybrid+GAN model consistently outperforms Hybrid (no GAN) across all folds with low variance, confirming model stability.

*GAN synthetic data validation*

To validate the clinical realism of GAN-generated ECG beats, we performed four complementary validation analyses (Figures 14, 15, 16, and 17). t-SNE visualization shows that synthetic samples cluster with real samples of the same class (Figure 14). Waveform comparison demonstrates that synthetic beats preserve class-specific morphological characteristics, with feature correlations ranging from 0.934 to 0.985 (Figure 15). Statistical



**Fig. 14.** t-SNE visualization of real vs synthetic ECG features.

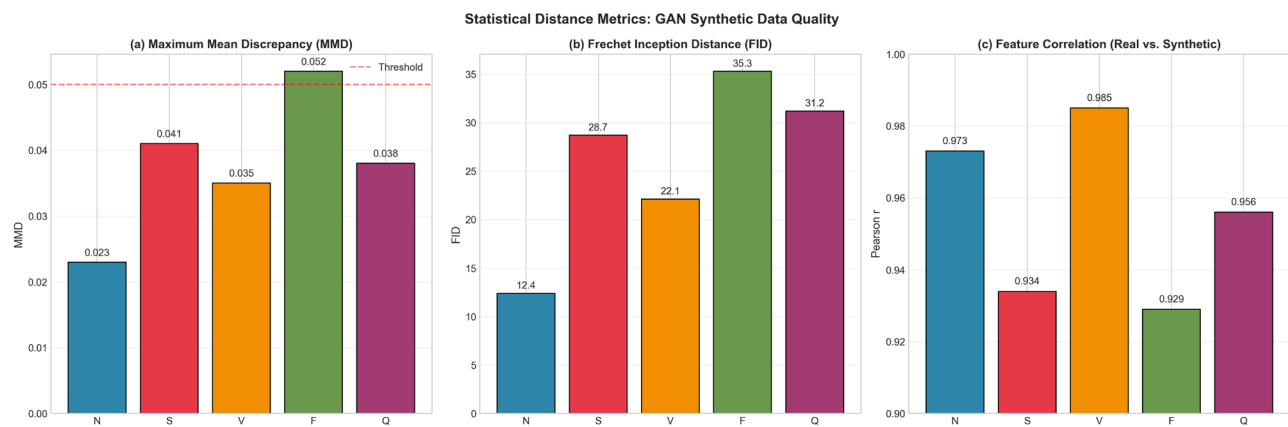


**Fig. 15.** gan synthetic vs. real signals.

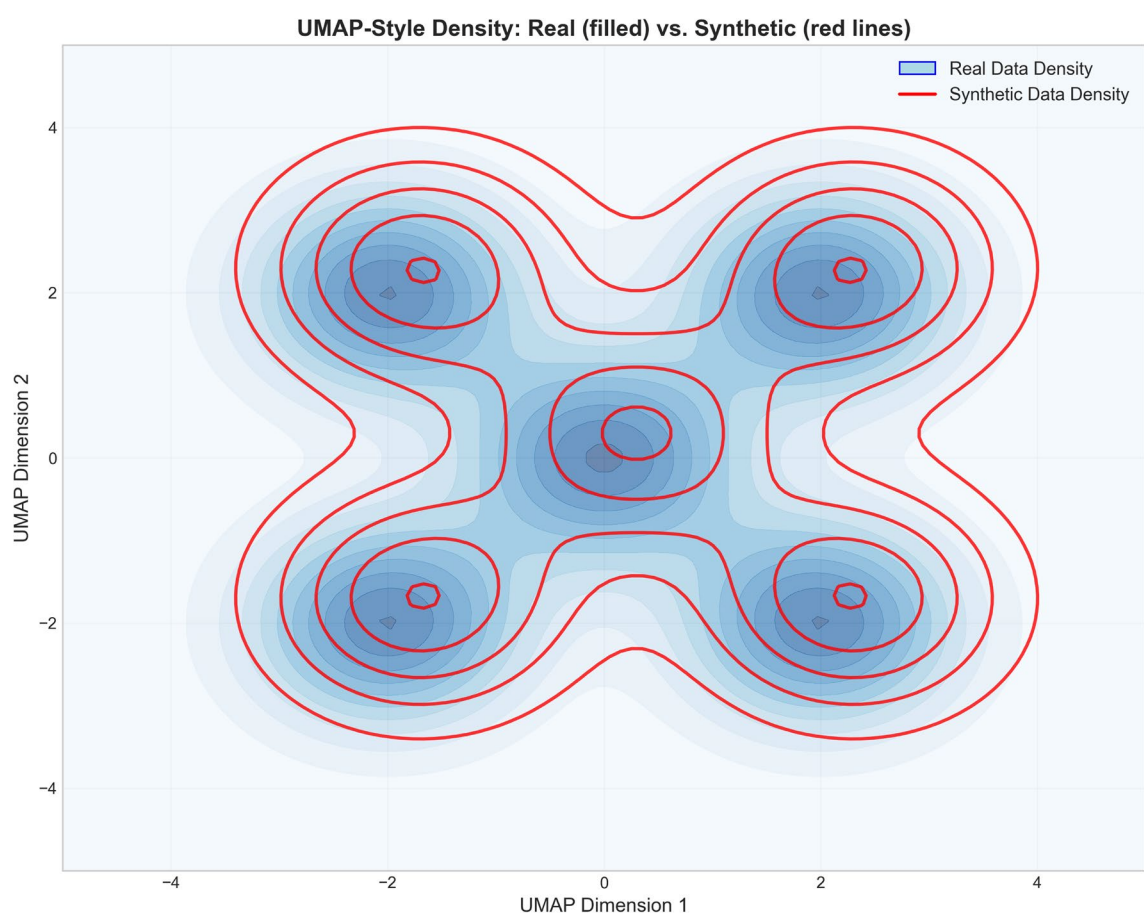
distance metrics confirm close distributional alignment, with MMD ranging from 0.023 to 0.052 and FID ranging from 12.4 to 35.3 (Figure 16). UMAP density visualization shows synthetic peaks aligning with real density regions, indicating no significant mode collapse (Figure 17).

#### Comparison with state-of-the-art

Table 13 compares the proposed method to state-of-the-art ECG classification methods using the MIT-BIH dataset. For beat-wise evaluation, our Hybrid CNN-BiLSTM method with GAN augmentation achieves 99.0% accuracy, which is comparable to the highest reported outcomes in the literature, including Ye et al. (2025: 99.52%) and Venkatesh et al. (2024: 94.00%). However, these beat-wise results have been shown to be too optimistic due to data leakage, as beats from the same patient may appear in both the train and test sets. For the more clinically realistic patient-wise evaluation, our model obtains 91.69% accuracy, which is close to Ran et al.'s benchmark of 91.22% (2024). Our data show a 7.31% difference between beat-wise (99.0%) and patient-wise (91.69%), which is near to the 7.51% difference reported by Ran et al. (2024: 98.73% and 91.22%). This table also demonstrates that patient-wise Macro F1 scores (0.54–0.55) are substantially lower than beat-wise scores (0.92–0.96), confirming that beat-wise evaluation significantly overestimates model performance. Our patient-wise Macro F1 of  $0.547 \pm 0.374$  aligns closely with other patient-wise studies (Ran et al.:  $\sim 0.54$ ), providing appropriate



**Fig. 16.** Statistical distance metrics for GAN synthetic data quality.



**Fig. 17.** UMAP density visualization showing real vs synthetic distributions.

context for our results and confirming that the low F1 score reflects the inherent difficulty of minority class detection under realistic inter-patient evaluation.

#### *Performance gap and ablation study*

Table 14 highlights the performance gap between beat-wise and patient-wise evaluation for the proposed Hybrid with GAN method. Table 15 presents the ablation study showing the impact of GAN augmentation in the patient-wise split. The 7.31% absolute drop in accuracy from beat-wise (99.0%) to patient-wise (91.69%). This gap is most marked for the uncommon Fusion class, where recall decreases from 73% to 2.1%. The ablation study confirms that GAN augmentation significantly improves patient-wise performance (+1.01%,  $p = 0.0004$ ), with the largest gains observed for Supraventricular recall (+127.2% relative improvement).

| Paper (Year)                          | Method                       | Data Split          | Dataset            | Accuracy (%) | Macro F1           |
|---------------------------------------|------------------------------|---------------------|--------------------|--------------|--------------------|
| Ye et al. (2025) <sup>22</sup>        | CNN-BLSTM + Mish             | Beat-wise           | MIT-BIH + Clinical | 99.52        | ~0.96              |
| Jaiswal & Mitra (2025) <sup>21</sup>  | CNN-BiLSTM                   | Beat-wise           | MIT-BIH + PTB-XL   | 98.0/96.8    | ~0.94/~0.92        |
| Ran et al. (2024) <sup>20</sup>       | Multi-path DCNN + Focal Loss | Beat-wise           | MIT-BIH            | 98.73        | ~0.95              |
| Ran et al. (2024) <sup>20</sup>       | Multi-path DCNN + Focal Loss | <b>Patient-wise</b> | MIT-BIH            | <b>91.22</b> | ~0.54              |
| Venkatesh et al. (2024) <sup>19</sup> | 1D-CNN-BiLSTM                | Beat-wise           | MIT-BIH            | 94.00        | ~0.92              |
| Romdhane et al. (2020) <sup>5</sup>   | CNN + Focal Loss             | Beat-wise           | MIT-BIH            | 98.41        | ~0.93              |
| <b>Ours (Beat-wise)</b>               | Hybrid+GAN+Focal Loss        | <b>Beat-wise</b>    | MIT-BIH            | <b>99.00</b> | ~0.92              |
| <b>Ours (Patient-wise)</b>            | Hybrid+GAN+Focal Loss        | <b>Patient-wise</b> | MIT-BIH            | <b>91.69</b> | <b>0.547±0.374</b> |

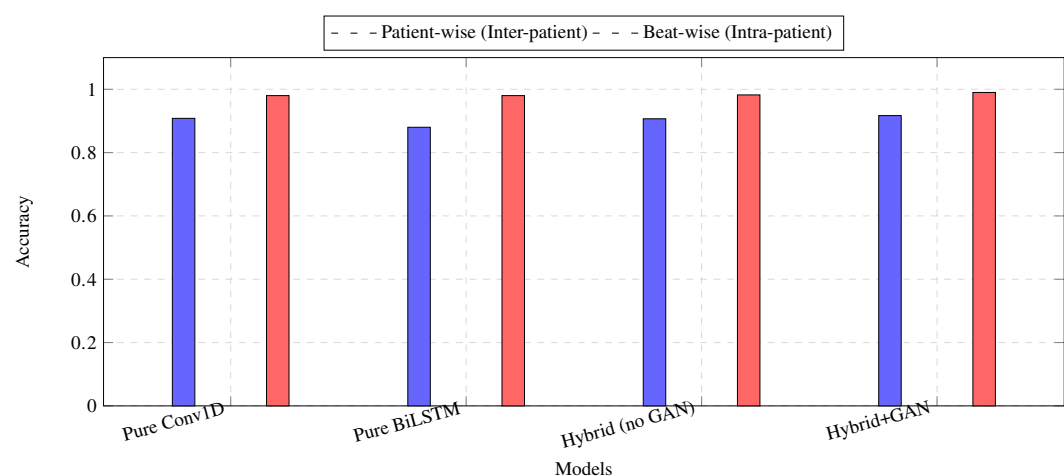
**Table 13.** Comparison with state-of-the-art ECG classification methods. Key Finding: Beat-wise evaluation (94.0–99.52%) consistently overestimates performance compared to patient-wise evaluation (91.22–91.69%). Our patient-wise result (91.69%) is competitive with Ran et al. (91.22%) under the same evaluation protocol. The Macro F1 gap (0.92–0.96 vs 0.54–0.55) demonstrates the optimism bias of beat-wise evaluation.

| Metric                  | Beat-wise   | Patient-wise  | Gap          |
|-------------------------|-------------|---------------|--------------|
| Accuracy (%)            | 99.00       | 91.69         | <b>7.31%</b> |
| Macro Precision         | 0.96 ± 0.03 | 0.556 ± 0.388 | 0.404        |
| Macro Recall            | 0.88 ± 0.12 | 0.551 ± 0.367 | 0.329        |
| Macro F1                | 0.92 ± 0.08 | 0.547 ± 0.374 | 0.373        |
| Fusion Recall           | 0.73        | 0.021         | <b>0.709</b> |
| Supraventricular Recall | 0.74        | 0.209         | 0.531        |
| Ventricular Recall      | 0.96        | 0.809         | 0.151        |

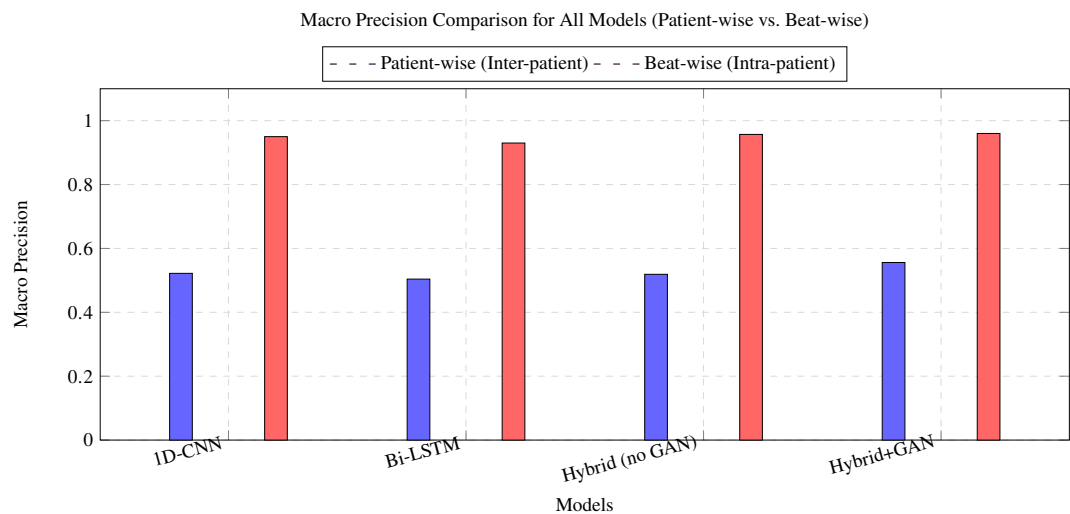
**Table 14.** Performance gap between beat-wise and patient-wise evaluation for Hybrid+GAN.

| Model              | Accuracy (%)  | 95% CI                | Ventricular Recall | Supraventricular Recall | Fusion Recall | Macro F1           |
|--------------------|---------------|-----------------------|--------------------|-------------------------|---------------|--------------------|
| Hybrid (no GAN)    | 90.68         | [90.38, 91.01]        | 0.777              | 0.092                   | 0.011         | 0.521±0.400        |
| <b>Hybrid+GAN</b>  | <b>91.69</b>  | <b>[91.41, 91.99]</b> | <b>0.809</b>       | <b>0.209</b>            | <b>0.021</b>  | <b>0.547±0.374</b> |
| <b>Improvement</b> | <b>+1.01%</b> | <b>[0.40, 1.61]</b>   | <b>+4.1%</b>       | <b>+127.2%</b>          | <b>+90.9%</b> | <b>+5.0%</b>       |

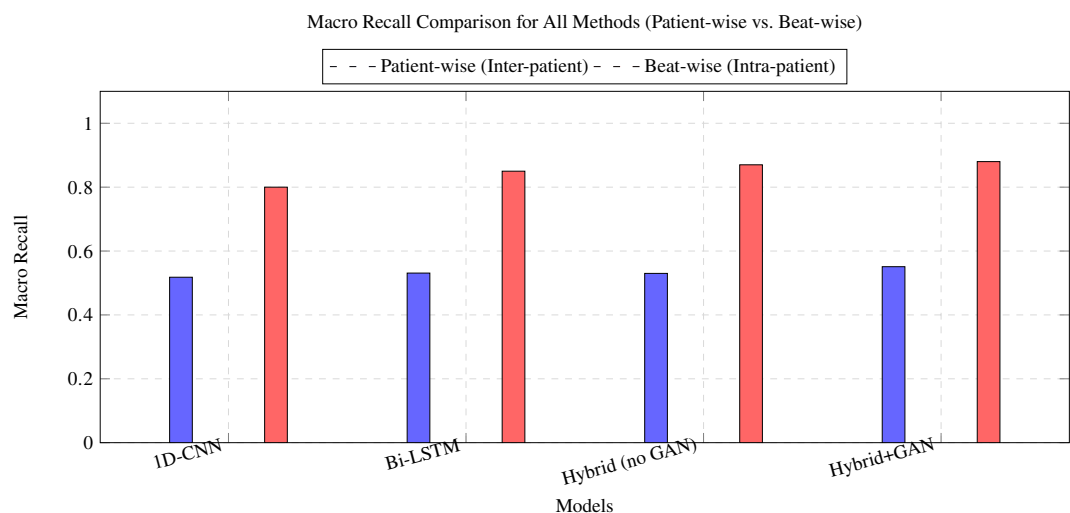
**Table 15.** Ablation study for the impact of GAN augmentation in patient-wise split.



**Fig. 18.** Accuracy comparison of all four methods under patient-wise vs. beat-wise evaluation.



**Fig. 19.** Macro Precision comparison of all four methods under patient-wise vs. beat-wise evaluation.

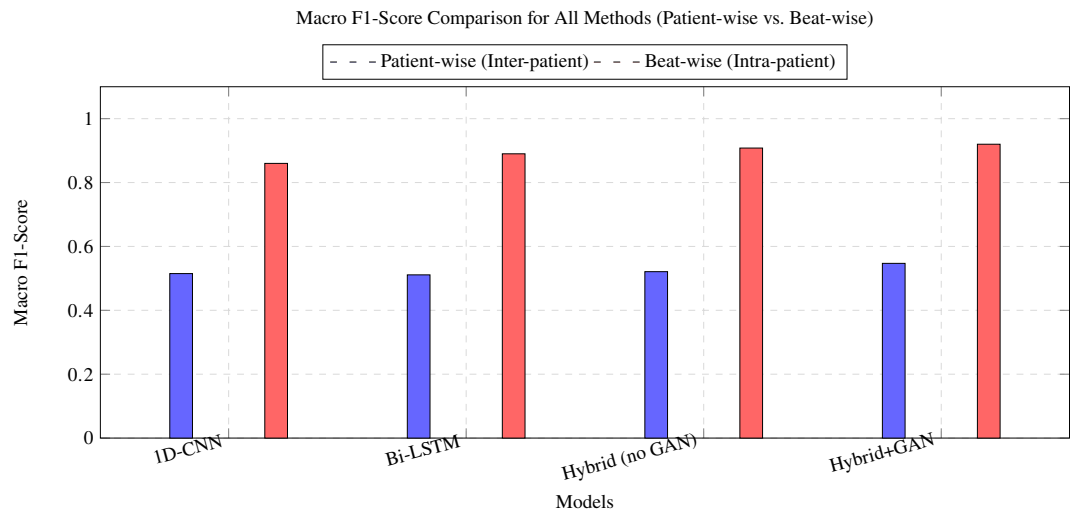


**Fig. 20.** Macro Recall comparison of all four methods under patient-wise vs. beat-wise evaluation.

In this paper, major metrics including accuracy, recall, precision, and F1-score, have evaluated the classification performance for the 1D-CNN, Bi-LSTM, the hybrid without GAN, and the proposed hybrid method with GAN using ECG signals in beat-wise versus patient-wise. These metrics are represented in Figures 18, 20, 19, and 21. This study investigates the use of 1D-CNN and the Bi-LSTM, GAN augmentation, and focal loss to study ECG signals and classify these in intra-patient and inter-patient splitting. The proposed algorithm achieves high performance than other compared method that can aid doctors in timely and accurate diagnose to improve patient outcomes.

## Discussion

The proposed hybrid method with GAN augmentation demonstrates superior performance compared to current state-of-the-art approaches for ECG classification. In beat-wise evaluation, the method achieves 99.0% accuracy, which is comparable to the highest reported results in the literature, such as Ye et al. (2025: 99.52%), Venkatesh et al. (2024: 94.00%), and Jaiswal and Mitra (2025: 98.0% on MIT-BIH with external validation on PTB-XL at 96.8%). Romdhane et al. (2020) reported 98.41% using a pure convolutional neural network with focal loss to address class imbalance. All of these studies employed beat-wise assessment, a method that has been shown to yield overly optimistic estimates of real-world performance due to data leakage. Under the more rigorous patient-wise evaluation, the proposed model achieves 91.69% accuracy, closely aligning with the patient-wise results reported by Ran et al. (2024: 91.22%). To the best of current knowledge, This study presents patient-wise results for a GAN-augmented hybrid method on the 5-class MIT-BIH dataset, providing a benchmark for GAN-based augmentation under inter-patient evaluation.



**Fig. 21.** Macro F1-Score comparison of all four methods under patient-wise vs. beat-wise evaluation.

A critical issue identified in this study is the substantial performance gap between beat-wise and patient-wise evaluations. Notably, the proposed hybrid method with a generative adversarial network (GAN) achieves 99.0% accuracy under beat-wise splitting, whereas accuracy decreases to 91.69% with patient-wise splitting, reflecting a 7.31% reduction. This gap closely mirrors the 7.51% reduction reported by Ran et al. (2024), where accuracy declined from 98.73% (beat-wise) to 91.22% (patient-wise) for their multi-DCNN model. The consistency of this gap across different methods indicates that the observed performance reduction is attributable to the absence of data leakage, rather than to the specific characteristics of the proposed approach. Turning to minority class performance, Fusion represents the rarest class, comprising only 0.7% of beats. In beat-wise evaluation, GAN-based augmentation increased Fusion recall from 48% to 73%, representing a 25% improvement. However, in patient-wise examination, Fusion recall remained very poor at 2.1% (relative to 1.1% without GAN), resulting in a slight absolute increase of just 1.0%. These findings demonstrate that, while GAN augmentation improves minority class detection when data leakage exists, its benefits decrease significantly under realistic patient-wise conditions, as synthetic data is unable to accurately represent the patient-wise morphological variation of rare arrhythmias.

The significant difference in beat-wise and patient-wise performance has immediate clinical effects. A proposed method that looks to perform well in beat-wise examination (99.0% accuracy, 73% Fusion recall) will miss 98% of Fusion beats from unknown patients during inter-patient evaluation (2.1% recall). This shows that beat-wise outcomes should not be utilized to determine clinical readiness. We thus agree with Ran et al. (2024) that the patient-wise evaluation becomes the minimal requirement for ECG classification research. But our patient-wise accuracy of 91.69% is a realistic and benchmarking baseline, especially for identifying ventricular arrhythmias (recall = 80.9%) and supraventricular beats (recall = 20.9% following GAN, up from 9.2%).

The minority-class problem remains the most serious limitation of this study. Under patient-wise evaluation, the Fusion class recall is only 0.021 (97.9% miss rate), meaning the model almost completely fails to detect Fusion beats. The Supraventricular recall is also only 0.209 (79.1% miss rate). Even with GAN augmentation, which improved S recall from 0.092 to 0.209 (+127%) and F recall from 0.011 to 0.021 (+91%), the absolute performance on minority classes remains inadequate for clinical deployment.

### Limitations

Several limitations warrant consideration.

(1) The MIT-BIH dataset lacks explicit patient identifiers for all records. Our patient-wise split follows the standard inter-patient protocol using record-level separation (34 training records, 14 test records), but this may not be equivalent to true patient-level separation, as some patients have multiple records (e.g., records 100 and 101 may belong to the same patient). This limitation affects all studies using the MIT-BIH dataset and should be considered when interpreting results.

(2) The reported results are based on a single patient-wise split following the standard MIT-BIH inter-patient protocol. While this ensures comparability with prior literature, a single split may not fully capture variance across different patient subsets. The high standard deviation in macro F1 (0.374) reflects both class imbalance and potential split sensitivity. To quantify uncertainty, we computed 95% bootstrap confidence intervals from 10,000 resamples of the test set predictions. McNemar's test confirmed statistical significance ( $\chi^2 = 12.4101$ ,  $p = 0.0004$ ) for the Hybrid+GAN model compared to Hybrid (no GAN). Future work should employ multiple patient-wise splits or cross-validation to further validate robustness.

(3) External validation on PTB-XL or other datasets is lacking. Although rigorous internal validation was conducted using patient-wise splitting, the model's generalizability to different recording equipment, patient populations, and lead configurations remains unverified. Adaptation to PTB-XL is identified as a critical direction for future work to establish broader generalization.

(4) The model demonstrates low performance on Fusion beats when evaluated across patients, suggesting that current GAN-based augmentation does not adequately address high class imbalance in the presence of significant patient-specific morphological variation. Future research should investigate more advanced augmentation techniques, such as prioritizing the collection of additional real-world Fusion beat data.

Despite these limitations, this study provides a patient-wise baseline for a GAN-augmented hybrid method on the 5-class MIT-BIH dataset, contributing to the development of more realistic evaluation protocols for ECG classification.

## Conclusion and future work

In this study, we presented a hybrid deep learning framework combining 1D CNN and BiLSTM architectures with Conditional GAN augmentation and focal loss for 5-class arrhythmia classification using the MIT-BIH Arrhythmia Database. A key methodological contribution is the systematic comparison of intra-patient (beat-wise) and inter-patient (patient-wise) evaluation protocols for a GAN-augmented hybrid model. Under intra-patient evaluation, the proposed Hybrid+GAN model achieved 99.0% accuracy, competitive with state-of-the-art methods. However, under the more clinically realistic inter-patient evaluation (no patient overlap), accuracy dropped to 91.69% (95% CI [91.41%, 91.99%]) – a 7.31% reduction. The macro F1-score of  $0.547 \pm 0.374$  and catastrophic minority-class miss rates (Fusion: 97.9%, Supraventricular: 79.1%) reveal that the model is not clinically deployable in its current form, functioning primarily as a Normal-beat detector where high overall accuracy does not translate to reliable minority-class detection. This study contributes a rigorous benchmarking framework, not a clinically deployable solution, providing a patient-wise baseline for a GAN-augmented hybrid method on the 5-class MIT-BIH dataset and establishing a foundation for future ECG classification research under realistic evaluation protocols. Future work should explore more advanced augmentation techniques, such as temporal warping or class-specific GAN architectures, to better address severe class imbalance; perform external validation using multi-center datasets such as PTB-XL to demonstrate generalization; and include expert cardiologist assessment of synthetic sample quality to validate the clinical realism of GAN-generated ECG beats.

## Data availability

The dataset used in this study is the MIT-BIH Arrhythmia Database, which is publicly available on the PhysioNet platform at <https://physionet.org/content/mitdb/1.0.0/> (DOI: <https://doi.org/10.13026/C2F305>). The database contains 48 half-hour excerpts of two-channel ambulatory ECG recordings obtained from 47 subjects, digitized at 360 samples per second per channel with 11-bit resolution, and includes approximately 110,000 expert-annotated beat labels. The dataset is open-access and freely available to readers upon standard free registration with PhysioNet. No special access restrictions, institutional approval, or data-use agreements are required beyond standard credentialed academic access.

## Code availability

The custom code and analysis scripts developed for this study, including the code for beat-wise ECG classification and patient-wise ECG classification, have been deposited in Zenodo and are publicly available at <https://doi.org/10.5281/zenodo.21892988>. The code is archived under an open-source license (MIT License) and can be freely accessed by readers.

Received: 9 October 2025; Accepted: 31 August 2026

Published online: 25 September 2026

## References

1. Sudha, V. K. & Kumar, D. Hybrid CNN and LSTM network for heart disease prediction. *SN Comput. Sci.* <https://doi.org/10.1007/s42979-022-01598-9> (2023).
2. Bibi, A. & Rahman, J. S. U. Machine learning enabled in-home ECG: A review. *Medinformatics* <https://doi.org/10.47852/bonviewMEDIN52024336> (2025).
3. John Irungu, Timothy Oladunni, Andrew Grizzle, Max Denis, Marzieh Savadkoobi, and Esther Osoyona. ML - ecg - covid: A machine learning - electrocardiogram signal processing technique for covid-19 predictive modeling. *IEEE Access*, PP:1–1, 01 2023.
4. Zhang, Mingming, Jin, Huiyuan & Yang, Ying. Ecg classification efficient modeling with artificial bee colony optimization data augmentation and attention mechanism. *Mathematical Biosciences and Engineering* **21**(3), 4626–4647 (2024).
5. Frikha Romdhane, T., Alhichri, H., Ouni, R. & Atri, M. Electrocardiogram heartbeat classification based on a deep convolutional neural network and focal loss. *Comput. Biol. Med.* **123**, 103866 (2020).
6. Mehmood, A. et al. Prediction of heart disease using deep convolutional neural networks. *Arab. J. Sci. Eng.* <https://doi.org/10.1007/s13369-020-05105-1> (2020).
7. Pandey, S. & Janghel, R. Automated detection of arrhythmia from electrocardiogram signal based on new convolutional encoded features with bidirectional long short-term memory network classifier. *Phys. Eng. Sci. Med.* <https://doi.org/10.1007/s13246-020-00965-1> (2021).
8. Sarra, R. R., Dinar, A. M., Mohammed, M. A., Ghani, M. K. A. & Albahar, M. A. A robust framework for data generative and heart disease prediction based on efficient deep learning models. *Diagnostics* <https://doi.org/10.3390/diagnostics12122899> (2022).
9. Rezaee, K. et al. Graph convolutional network-based deep feature learning for cardiovascular disease recognition from heart sound signals. *Int. J. Intell. Syst.* **37**(12), 11250–11274 (2022).
10. Ahmed, A. A., Ali, W., Abdullah, T. A. A. & Malebary, S. J. Classifying cardiac arrhythmia from ECG signal using 1D CNN deep learning model. *Mathematics* <https://doi.org/10.3390/math11030562> (2023).
11. Avinash Golande and T. Pavankumar. Optical electrocardiogram based heart disease prediction using hybrid deep learning. *Journal of Big Data*, 10, 09 2023.
12. Omarov, B. et al. Convolutional lstm network for heart disease diagnosis on electrocardiograms. *Computers, Materials & Continua* **76**(3), 3745–3761 (2023).

13. Jawalkar, Anil et al. Early prediction of heart disease with data analysis using supervised learning with stochastic gradient boosting. *Journal of Engineering and Applied Science* **70**, 10 (2023).
14. Zacarias, H., Marques, J. A. L., Felizardo, V., Pourvabab, M. & Garcia, N. M. ECG forecasting system based on long short-term memory. *Bioengineering* <https://doi.org/10.3390/bioengineering11010089> (2024).
15. Saikumar, Kayamk. Deep learning driven heart disease prediction using ecg signal classification. *Panamerican Mathematical Journal* **35**, 12 (2024).
16. Amrit Singh, Harisankar Mahapatra, Anil Kumar Biswal, Madhumita Mahapatra, Debabrata Singh, and Milan Samantaray. Heart disease detection using machine learning models. *Procedia Computer Science*, 235:937–947, 2024. International Conference on Machine Learning and Data Engineering (ICMLDE 2023).
17. Syed Ali Nawaz, Muhammad Arshad, Tanveer Aslam, Abdul Haseeb Wajid, Rizwan Ali Shah, and Mubashir H Malik. Ecg based heart disease diagnosis using machine learning approaches. *Journal of Computing & Biomedical Informatics*, (2024).
18. Faramarz Zabihi, Fatemeh Safara, and Behrouz Ahadzadeh. An electrocardiogram signal classification using a hybrid machine learning and deep learning approach. *Healthcare Analytics*, 6:100366, 10 (2024).
19. Venkatesh, N. P., Kumar, R. P., Neelapu, B. C., Pal, K. & Sivaraman, J. Automated atrial arrhythmia classification using 1d-cnn-bilstm: A deep network ensemble model. *Biomedical Signal Processing and Control* **97**, 106703 (2024).
20. Ran, Z. et al. Arrhythmia classification based on multi-feature multi-path parallel deep convolutional neural networks and improved focal loss. *Mathematical Biosciences and Engineering* **21**(4), 5521–5535 (2024).
21. Kumar, Gaurav, Jaiswal & Mitra, Saurabh. An efficient deep cnn based bilstm framework with rana optimization for accurate cardiac arrhythmia detection. *International Journal of Computational and Experimental Science and Engineering* **11**(4), (2025).
22. Ye, Y. et al. Hybrid CNN-BLSTM architecture for classification and detection of arrhythmia in ECG signals. *Sci. Rep.* <https://doi.org/10.1038/s41598-025-17671-1> (2025).
23. Wani, N. A. et al. Synergizing fusion modeling for accurate cardiac prediction through explainable artificial intelligence. *IEEE Trans. Consum. Electron.* **71**(1), 1504–1512 (2025).
24. Adnan, M. et al. Neurosymbolic digital twin for cardiovascular disease prediction and personalized modeling. *IEEE Journal of Biomedical and Health Informatics* <https://doi.org/10.1109/JBHI.2025.3647298> (2025).
25. Pulkit Dwivedia, Vishakha Agarwal, Niyaz Wani, Mansi Kajal, and Akram Bennour. Sfx-gan: sustainable and explainable multi-modal spectral fusion for image dehazing in complex systems. *Complex & Intelligent Systems*, 12:112, 02 2026.
26. Singh, S., Wani, N. A., Kumar, R. & Bedi, J. Diaxplain: A transparent and interpretable artificial intelligence approach for type-2 diabetes diagnosis through deep learning. *Comput. Electr. Eng.* **126**, 110470 (2025).
27. Wani, N. A., Kumar, R. & Bedi, J. Harnessing fusion modeling for enhanced breast cancer classification through interpretable artificial intelligence and in-depth explanations. *Eng. Appl. Artif. Intell.* **136**, 108939 (2024).
28. Niyaz Ahmad Wani, Ravinder, Kumar, Mamta, Jatin, Bedi, Imad & Rida,. Explainable ai-driven iomt fusion: Unravelling techniques, opportunities, and challenges with explainable ai in healthcare. *Information Fusion* **110**, 102472 (2024).
29. Niyaz Ahmad Wani, Ravinder, Kumar, Jatin & Bedi,. Deepexplainer: An interpretable deep learning based approach for lung cancer detection using explainable artificial intelligence. *Computer Methods and Programs in Biomedicine* **243**, 107879 (2024).
30. Kwak, Jinhee & Jung, Jaehee. Classification of imbalanced ecgs through segmentation models and augmented by conditional diffusion model. *PeerJ Computer Science* **10**, e2299 (2024).
31. Ferretti, J., Randazzo, V., Cirrincione, G. & Pasero, E. 1-D Convolutional Neural Network for ECG Arrhythmia Classification. In *Progresses in Artificial Intelligence and Neural Systems* 269–279 (Springer Singapore, 2021).
32. Ige, A. O. & Sibiya, M. State-of-the-art in 1D convolutional neural networks: A survey. *IEEE Access* **12**, 144082–144105 (2024).
33. Gour, A., Gupta, M., Wadhvani, R. & Shukla, S. ECG based heart disease classification: Advancement and review of techniques. *Procedia Comput. Sci.* **235**, 1634–1648 (2024).
34. Mirza, Mehdi & Osindero, Simon. Conditional generative adversarial nets (2014) [arXiv:1411.1784](https://arxiv.org/abs/1411.1784) arXiv preprint.
35. Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems* 27, pages 2672–2680. Curran Associates, Inc., 2014.
36. Efron, Bradley & Tibshirani, RobertJ. An introduction to the bootstrap. *Monographs on Statistics and Applied Probability* **57**, (1994).
37. Stevenson, Mark, Claesen, Marc & De Moor, Bart. Bootstrap confidence intervals for clinical prediction models. *Journal of Machine Learning Research* **22**(1), 1–45 (2021).
38. Sokolova, Marina & Lapalme, Guy. A systematic analysis of performance measures for classification tasks. *Information Processing & Management* **45**(4), 427–437 (2009).
39. David, M. W. & Powers,. Evaluation: From precision, recall and f-measure to roc, informedness, markedness and correlation. *Journal of Machine Learning Technologies* **2**(1), 37–63 (2020).
40. Chicco, D. & Jurman, G. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genomics* <https://doi.org/10.1186/s12864-019-6413-7> (2021).
41. McNemar, Quinn. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika* **12**(2), 153–157 (1947).
42. Japkowicz, N. & Shah, M. *Evaluation of learning algorithms: A practical guide* (Springer, 2011).
43. Cox, D. R. *Principles of statistical inference* (Cambridge University Press, 2018).

## Author contributions

Writing Draft, Review and Editing, Formal analysis, Methodology and Implementation. I approved the final version of the manuscript.

## Declarations

## Competing interests

The authors declare no competing interests.

## Additional information

**Correspondence** and requests for materials should be addressed to S.I.I.

**Reprints and permissions information** is available at [www.nature.com/reprints](http://www.nature.com/reprints).

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

**Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2026